Rectilinear Motion

Kinematics: It is branch of physics which deals with motion without, going into details about cause of motion.

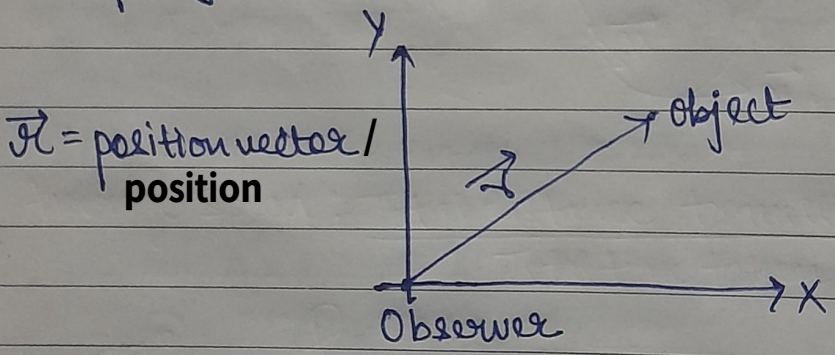
Rest and motion:

→ They are property of object as well as observer

→ If position of object wrt. observer remains unchanged then object is at rest wrt observer.

→ If position of object wrt observer is changing, then object is in motion wrt observer.

Position Vector: It is a vector which provides location of object wrt observer.



→ Vector quantity whose tail is at observer and head at object

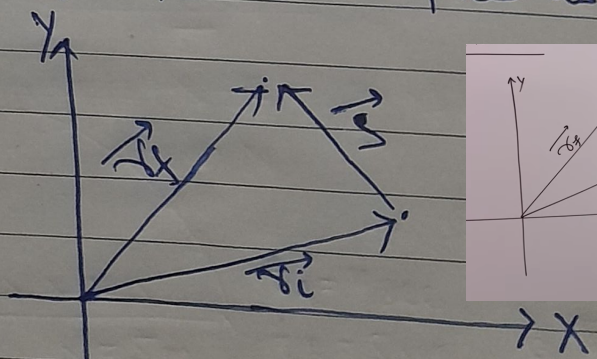
→ Unit: meter (m)

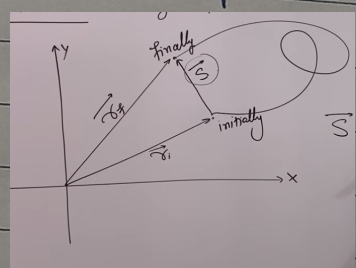
→ Dimension: [L]

Displacement: Change in position is defined as displacement.

$$\vec{S} = \vec{r}_f - \vec{r}_i$$

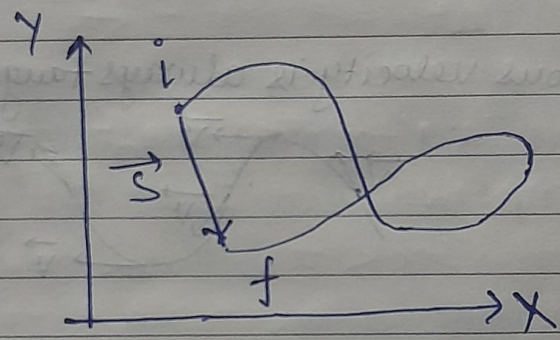
$$\vec{r}_f = \vec{S} + \vec{r}_i$$





→ Vector quantity whose direction is from initial to final position.
→ It does not depends on path.
→ It depends on initial and final positions only.
→ For a closed path (ie initial and final positions are same), $\vec{S}$ is a null vector or zero vector.
→ Magnitude of displacement ($|\vec{S}|$) = It is shortest distance or shortest separation b/w initial and final positions.

Distance : It is length of path travelled



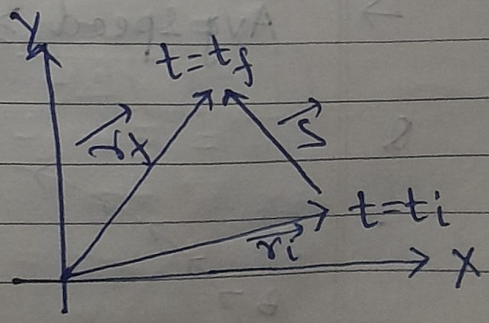
→ Scalar Quantity
→ unit = m Dimension = [L]
→ It depends on path
→ |Displacement| ≤ Distance Travelled

Equality holds only if motion is straight line without turning back.

Velocity : Rate of change of position is defined as velocity.

i) Average Velocity : defined for a finite time interval

$$\vec{V}_{avg} = \frac{\vec{r}_f - \vec{r}_i}{t_f - t_i}$$



Vector quantity ← $\boxed{\vec{V}_{avg} = \frac{\vec{S}}{\Delta t}}$

unit = m/s
Dimension = $[LT^{-1}]$

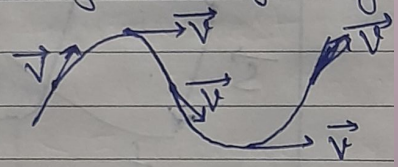
→ Direction of $\vec{v}_{avg.}$ is along $\vec{s}$.
→ In a closed path $\vec{v}_{avg.}$ velocity is null vector.
→ It is defined for finite time interval.

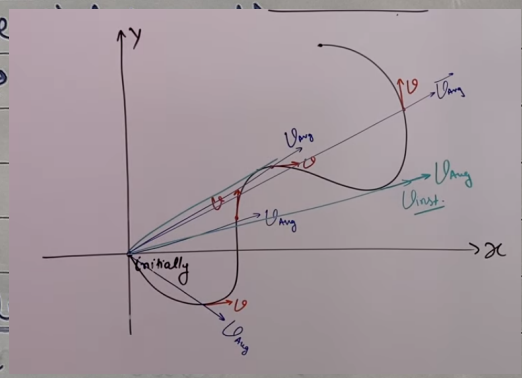
ii) Instantaneous Velocity: Velocity at an instant.

$$\vec{v}_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt}$$

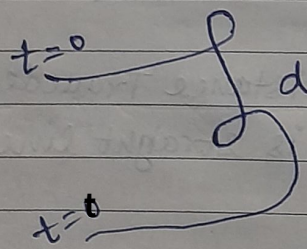
If particle is moving only along X-axis $\Rightarrow$ $v_{inst} = \frac{dx}{dt}$

→ Unit = m/s
→ Dimension = $[LT^{-1}]$
→ Vector
→ Instantaneous velocity is always tange





Average Speed: $\frac{\text{Total distance travell}}{\text{Total time taken}}$



$$v = \frac{d}{t}$$

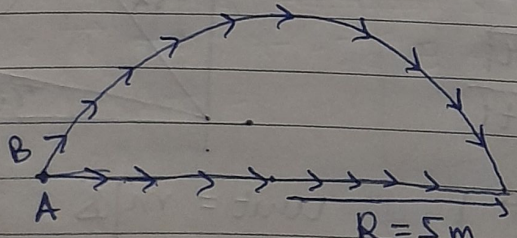
Unit = m/s or km/hr

$$1\,km/hr = \frac{5}{18}\,m/s$$

Dimension = $[LT^{-1}]$

→ Scalar Quantity
→ Avg speed $\geq$ |Avg velocity|

Q



Find: ① displacement of A and B
② Distance travelled by A and B
③ v_{avg} of A and B
④ v_{avg} of A and B (speed)

① Displacement of A = 10m
Displacement of B = 10m

② Distance Travelled by A = 10m
Distance Travelled by B = 5π m = 15.7m

③ Avg Velocity of A = 10/4 = 5/2 m/s = 2.5 m/s
Avg Velocity of B = 10/5 = 2 m/s

④ Avg speed of A = 10/4 = 5/2 = 2.5 m/s
Avg speed of B = $5\pi/5 = \pi$ m/s = 3.14 m/s

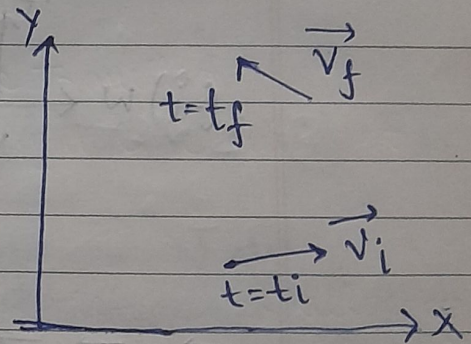
Instantaneous Speed: Speed at an instant.

→ Speed$_{ins} = |\vec{V}_{inst}|$ Dimension = $[LT^{-1}]$
→ Unit = m/s
→ Scalar

Acceleration: Rate of change of velocity.

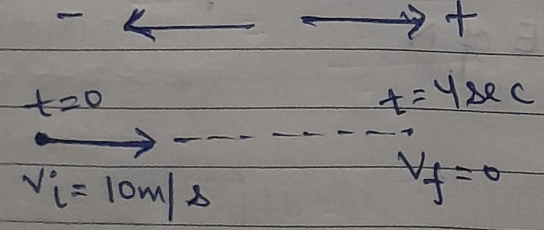
i) Average Acceleration:
Defined for finite time interval.

$$\vec{a}_{avg} = \frac{\vec{V}_f - \vec{V}_i}{t_f - t_i} = \frac{\Delta\vec{V}}{\Delta t}$$



→ Unit = m/s^2
→ Dimension = $[LT^{-2}]$
→ Vector Quantity
→ Direction is along change in velocity $(\vec{V}_f - \vec{V}_i)$

Q



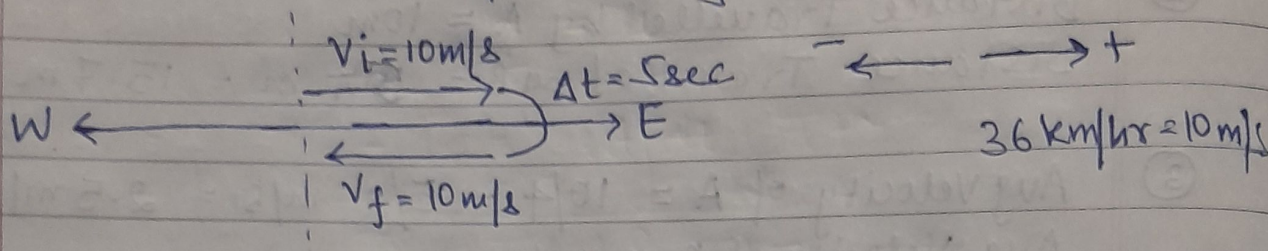
Find a_{avg} in 4 sec

Soln: $\vec{a}_{avg} = \frac{0-(+10)\,m/s}{4}$

$\vec{a}_{avg} = -2.5\,m/sec^2$

$\vec{a} = 2.5\,m/s$ (Retardation) / deceleration

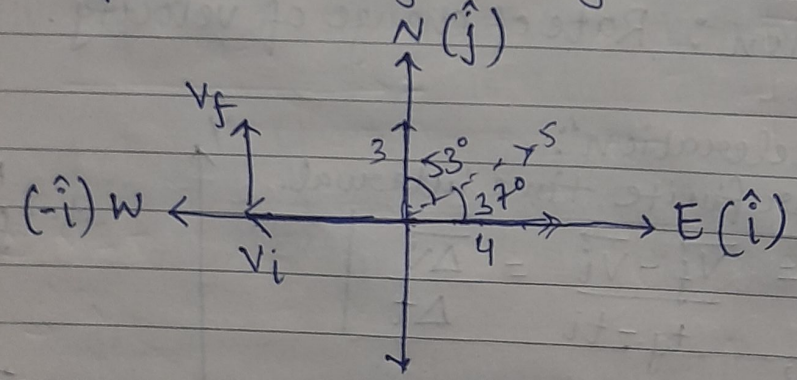
Q A car travelling along east with speed 36 km/hr takes a U-turn without changing speed. If turning took 5 sec, then find $\vec{a}_{avg}$ while turning.



$$\vec{a} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t} = \frac{-10-(+10)}{5} = \frac{-20}{5} = 4 \text{ m/s}^2 \text{ towards west}$$

Alternate $$\vec{a} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t} = \frac{10(-\hat{i}) - (10)\hat{i}}{5} = -4\hat{i} \text{ m/s}^2$$

Q A bike travelling towards west with speed 72 km/hr takes a right turn and speed changes to 54 km/hr. If this turn took 5 seconds, then find acceleration of bike during the right turn.



$v_i = 72$ km/hr $= 20$ m/s

$v_f = 54$ km/hr $= 15$ m/s

$$\vec{v}_i = -20\hat{i}$$

$$\vec{v}_f = 15\hat{j}$$

$$\vec{a} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t} = \frac{15\hat{j} - (-20\hat{i})}{5} = (4\hat{i} + 3\hat{j}) \text{ m/s}^2$$

$\vec{a} = 5 \text{ m/s}^2$ ~~towards N-E~~ 37° N of E

or $\vec{a} = 5 \text{ m/s}^2$ 53° E of N

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Instantaneous Acceleration: Acceleration at an instant

$$\vec{a}_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \qquad \vec{a} = \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right) = \frac{d^2\vec{r}}{dt^2}$$

x-axis ② $\rightarrow$ $v = \frac{dx}{dt} \rightarrow a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

Q $\vec{r} = (t^2\hat{i} + 5t\hat{j} + 3\hat{k})$ m

Find ① Velocity at any time t
② Velocity at $t = 5$ sec
③ Accn at any time t
④ Accn at $t = 5$ sec

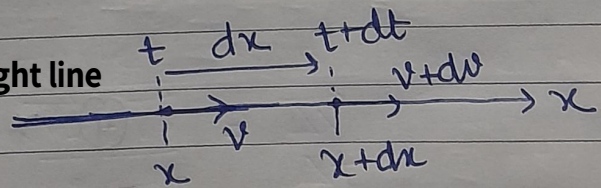
① $\vec{v} = \frac{d\vec{r}}{dt} = 2t\hat{i} + 5\hat{j}$ m/s

② $\vec{v}\big|_{t=5s} = 2(5)\hat{i} + 5\hat{j} = 10\hat{i} + 5\hat{j}$ m/s

③ $\vec{a} = \frac{d\vec{v}}{dt} = 2\hat{i}$ m/s^2 (const accn)

④ $\vec{a}\big|_{t=5sec} = 2\hat{i}$ m/s^2

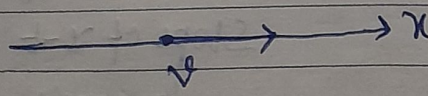
Note! only valid for straight line motion



$$a = \frac{dv}{dt} = \frac{dx}{dx} \times \frac{dv}{dt} = \frac{dx}{dt} \times \frac{dv}{dx} = v\frac{dv}{dx}$$

$$\boxed{a = v\frac{dv}{dx}}$$

Q



$V = \sqrt{10+8x}$

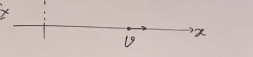
Find its accn.

Soln: $a = v\frac{dv}{dx}$

$= \sqrt{10+8x} \times \frac{8}{2\sqrt{10+8x}}$

$= 4$ m/s^2

Ex



$v = \sqrt{10+8x}$

Find its acceleration

$v^2 = 10 + 8x$

$\frac{d}{dx}$

$2v\frac{dv}{dx} = 0 + 8(1)$

$a = \frac{v\,dv}{dx} = 4$ m/s^2

Rectilinear Motion
→ Straight line
1 - Dimensional motion

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Uniform motion : Constant velocity / Zero acceleration

$u = \text{const}$

$$\frac{dx}{dt} = u$$

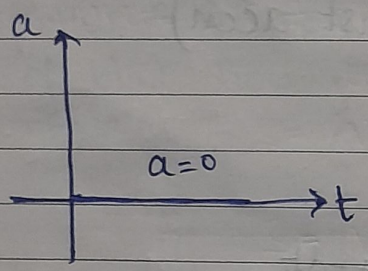
$$\int_{x_i}^{x_f} dx = u \int_0^t dt$$

$$\Rightarrow x_f = x_i + ut \qquad [S = x_f - x_i]$$

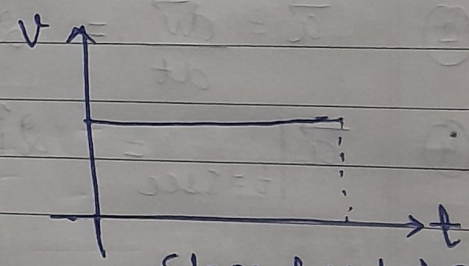
$$\Rightarrow \boxed{S = ut} \qquad (S = \text{distance} = \text{displacement})$$

Graphs :

① a v/s t

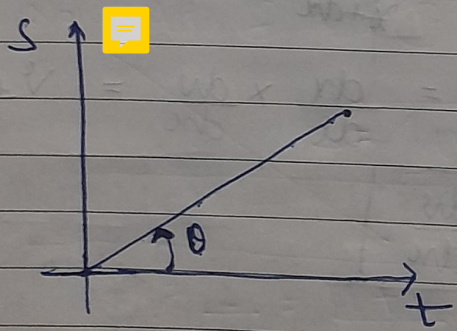


② ~~x~~ v v/s t



Slope of v-t ⇒ accn

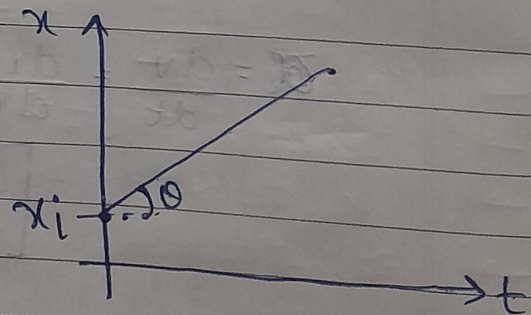
③ S v/s t



$\tan\theta = u$

Slope of S-t ⇒ u (velocity)
(S = displacement)

④ x v/s t

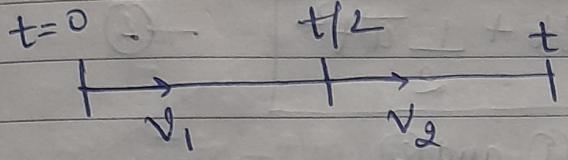


$x = x_i + ut$

Slope of x-t ⇒ velocity
(x = position)

→ Area under a-t curve ⇒ Change in velocity
→ Area under v-t curve ⇒ Displacement
→ Area under speed-t curve ⇒ Distance

Q A boy running on straight line runs with velocity v_1 for half of the time and with velocity v_2 for rest half of the time. Find v_{avg} for entire path.

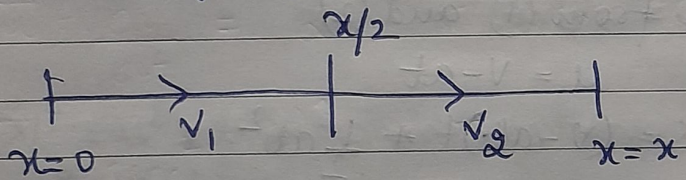


$$v_{avg} = \frac{s}{t} = \frac{v_1\frac{t}{2} + v_2\frac{t}{2}}{t}$$

$\leftarrow v_1\frac{t}{2} \rightarrow \leftarrow v_2\frac{t}{2} \rightarrow$

$$v_{avg} = \frac{v_1 + v_2}{2}$$

Q A boy running on straight line runs with velocity v_1 for half of the distance and with velocity v_2 for rest half of the distance. Find v_{avg}.



$$v_{avg} = \frac{x}{t_1 + t_2}$$

$$= \frac{x}{\frac{x}{2v_1} + \frac{x}{2v_2}}$$

$$v_{avg} = \frac{2v_1v_2}{v_1 + v_2}$$

Uniformly Accelerated Motion: $\vec{a} = \text{const}$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$= \int_{\vec{u}}^{\vec{v}} d\vec{v} = \vec{a}\int_0^t dt$$

$$= \vec{v} - \vec{u} = \vec{a}t$$

$$= \boxed{\vec{v} = \vec{u} + \vec{a}t}$$

$$\vec{v} = \vec{u} + \vec{a}t = \frac{d\vec{r}}{dt}$$

$$\Rightarrow \int_{\vec{r}_i}^{\vec{r}_f} d\vec{r} = \vec{u}\int_0^t dt + \vec{a}\int_0^t t\,dt$$

$$= \vec{r}_f - \vec{r}_i = \vec{u}t + \frac{1}{2}\vec{a}t^2$$

$$= \boxed{\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2}$$

$v = u + at$ — (1)

$s = ut + \frac{1}{2}at^2$ — (2)

Eliminate t from (1) and (2)

$$s = u\left(\frac{v-u}{a}\right) + \frac{1}{2}a\left(\frac{v-u}{a}\right)^2$$

$$\Rightarrow \boxed{v^2 = u^2 + 2as} \quad \text{——} \quad (3)$$

Eliminate u from (1) and (2)

$$u = v - at$$

$$s = (v-at)t + \frac{1}{2}at^2$$

$$\boxed{s = vt - \frac{1}{2}at^2} \quad \text{——} \quad (4)$$

Now Add (2) and (4)

$$\boxed{s = \frac{(u+v)t}{2}}$$

$$\boxed{v_{avg} = \frac{s}{t} = \frac{u+v}{2}}$$

$v = u + at$ — (1)

$s = ut + \frac{1}{2}at^2$ — (2)

$v^2 = u^2 + 2as$ — (3)

$s = vt - \frac{1}{2}at^2$ — (4)

$s = \frac{(u+v)}{2}t$ — (5)

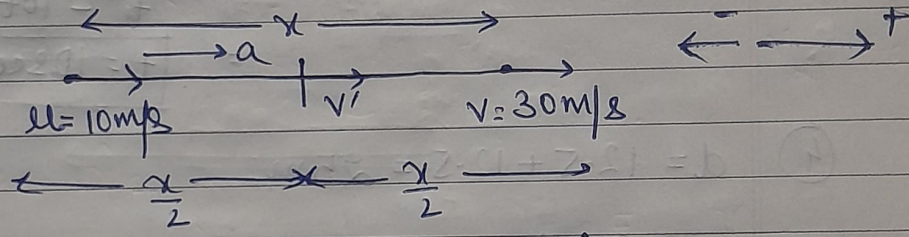
$v_{avg} = \frac{s}{t} = \frac{u+v}{2}$ — (6)

Displacement in n^{th} second (Time interval of 1 sec from $(n-1)$ s to n sec)

$$S_n = S(t=n) - S(t=n-1)$$
$$= un + \frac{1}{2}an^2 - \left(u(n-1) + \frac{1}{2}a(n-1)^2\right)$$

$$\boxed{S_n = u + \frac{a}{2}(2n-1)}$$

Q A particle moving along a straight line with uniform acceleration has initial velocity 10 m/s. After travelling some distance velocity becomes 30 m/s. Find velocity of particle when it is at mid point of the path.



$v^2 = u^2 + 2ax$

$30^2 = 10^2 + 2ax$

$2ax = 800$

$\boxed{ax = 400}$

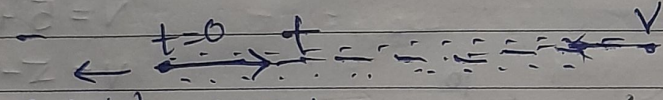
Now, $v'^2 = u^2 + 2a\frac{x}{2}$

$= v'^2 = u^2 + ax$

$= v'^2 = 100 + 400$

$= v' = \sqrt{500} = 10\sqrt{5}$ m/s

Q



① Find the time when particle stops momentarily.
② Find the displacement of particle till it stops momentarily.
③ Find the time when particle reaches its initial pos.
④ Find distance travelled by the particle till it reaches its initial position.
⑤ Draw a-t curve
⑥ Draw v-t curve
⑦ Draw speed-t curve
⑧ Draw s-t curve
⑨ Draw distance-t curve

Soln ①

$u = +5\ m/s$

$a = -1\ m/s^2$

$v = 0$

$v = u + at$

$0 = 5 - 1(t)$

$\boxed{t = 5\ sec}$

②

$s = ut + \frac{1}{2}at^2$

$s = 5(5) + \frac{1}{2}(-1)(5)^2 = 25 - 12.5$

$s = 12.5\ m$

③

$s = 0$

$u = +5\ m/s$

$a = -1\ m/s^2$

$s = ut + \frac{1}{2}at^2$

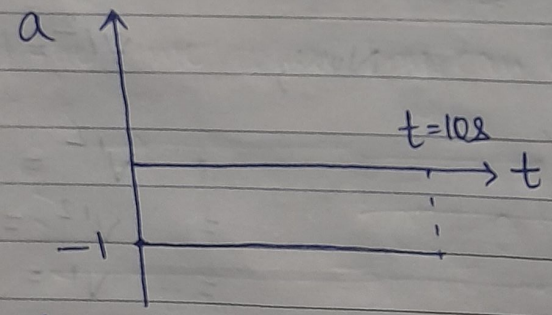
$= 0 = 5t + \frac{1}{2}t^2$

$= t^2 - 10t = 0$

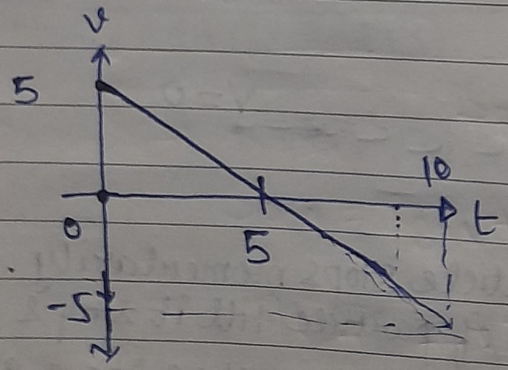
$= \boxed{t = 10\ sec}$

④ $d = 12.5 + 12.5 = 25\ m$

⑤



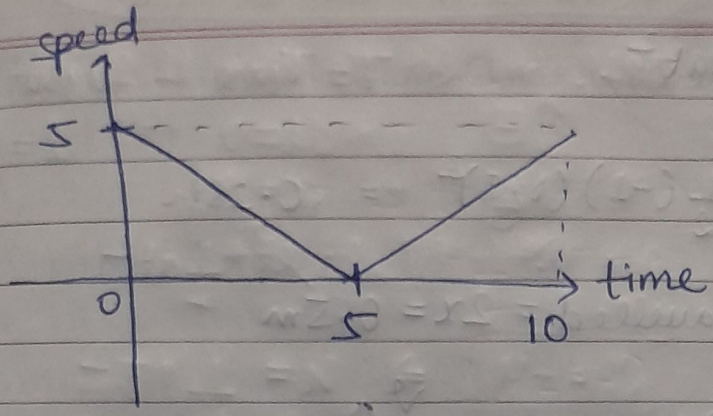
⑥



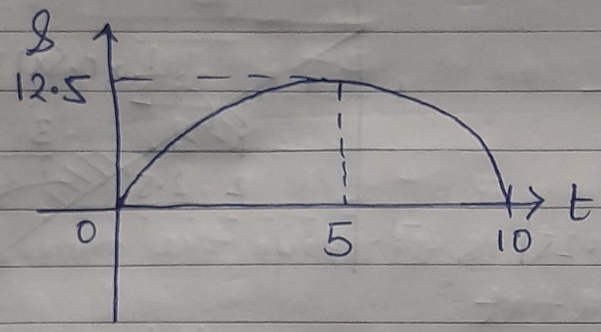
$v = 5 + (-1)t$

$v = 5 - t$

(7)



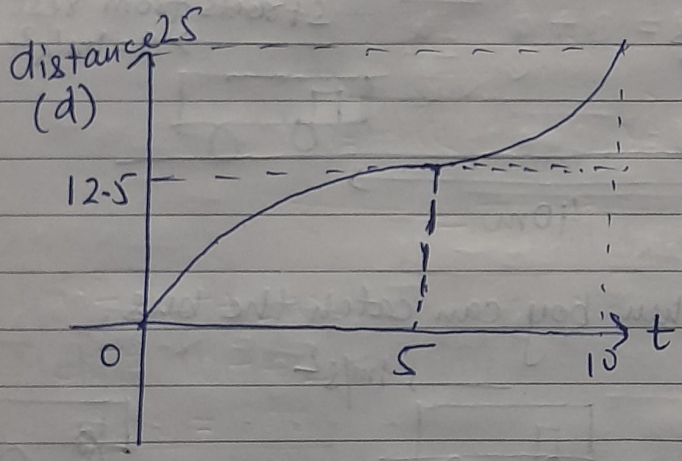
(8)



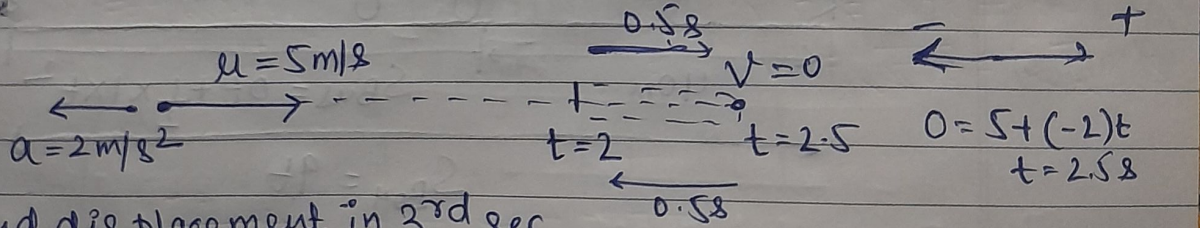
$$s = 5t + \frac{1}{2}(-1)t^2$$

$$\boxed{s = 5t - \frac{1}{2}t^2}$$

(9)



Q



(1) Find displacement in 3rd sec

(2) Find distance travelled in 3rd second

Soln (1)

$u = +5\,m/s$

$a = -2\,m/s^2$

$$S_n = u + \frac{a}{2}(2n-1)$$

$$S_3 = 5 - \frac{2}{2}(2(3)-1)$$

$$\boxed{S_3 = 0}$$

(2) $x = vt - \frac{1}{2}at^2$

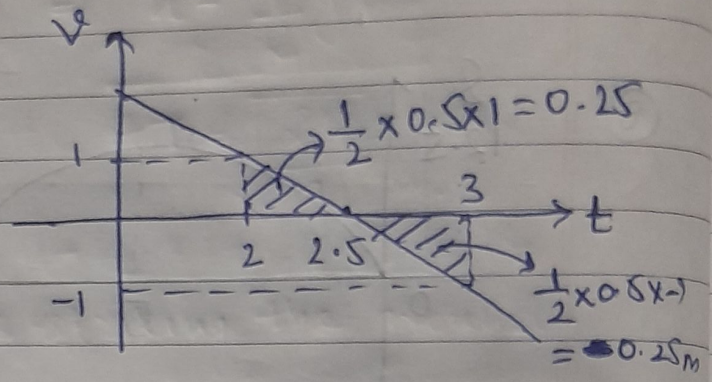
$x = 0 - \frac{1}{2}(-2)(0.5)^2 = 0.25m$

Distance travelled $= 2x = 0.5m$

Alternate

$V = 5 + (-2)t$

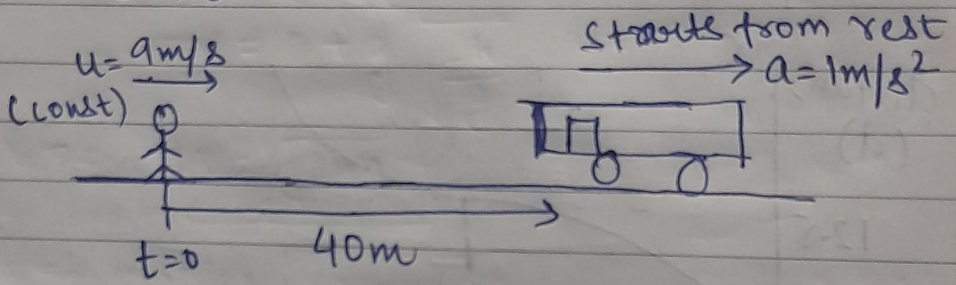
$V = 5 - 2t$



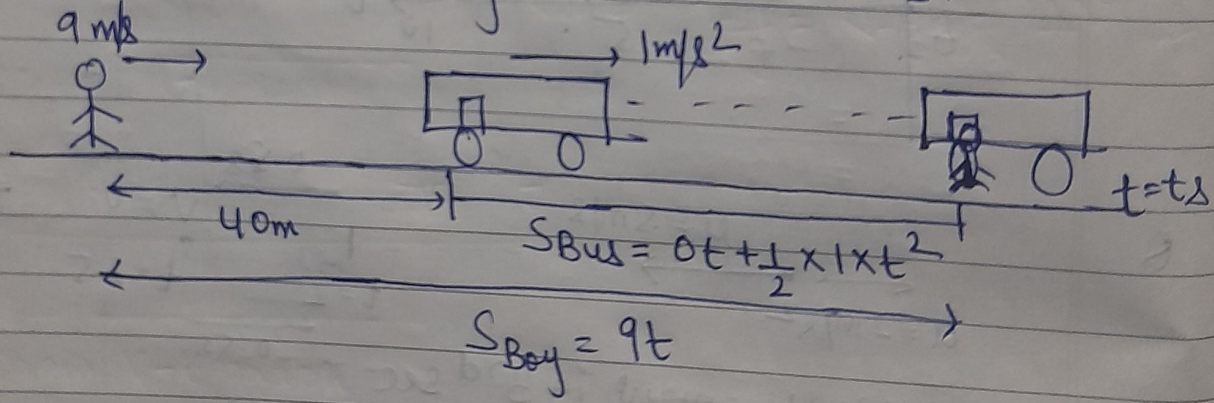
$d = 0.25 + |-0.25|$

$d = 0.50m$

Q



Find the time when boy can catch the bus.



$S_{Boy} = S_{Bus} + 40$

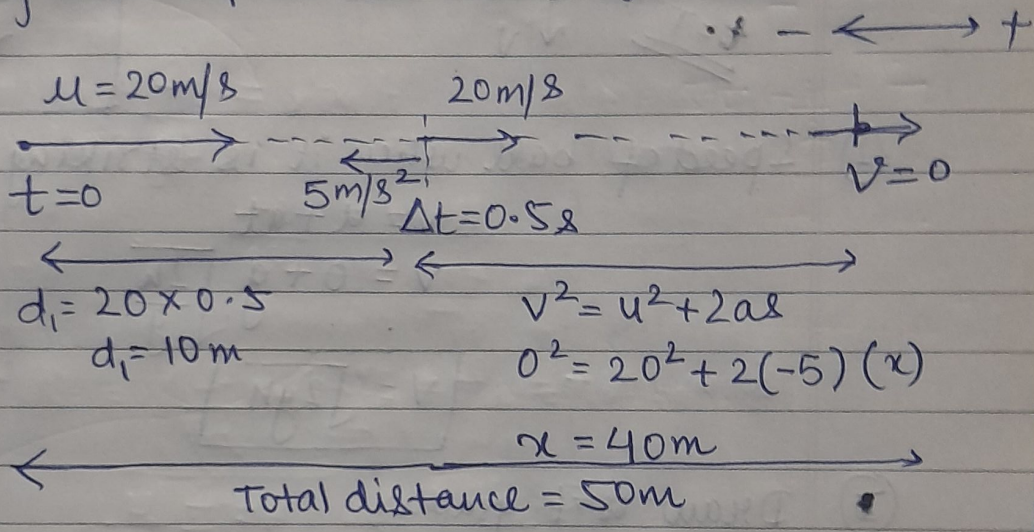
$= 9t = \frac{1}{2}t^2 + 40$

$= t^2 - 18t + 80 = 0$

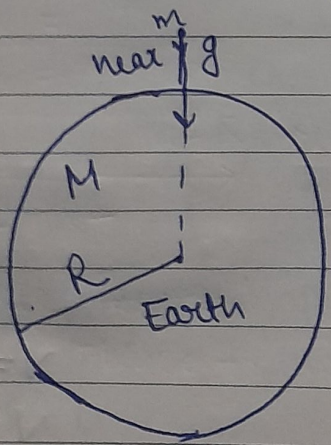
$\boxed{t = 8s \text{ or } t = 10s}$

Reaction Time : Time interval to observe, analyze and react.

Q A biker has reaction time $\Delta t = 0.5s$. He is riding a bike with constant velocity 72km/hr. At a time he observes an obstacle in the path and applies brakes. Brakes provides retardation of $5 m/s^2$. Find distance travelled by bike after the biker observes the obstacle.



Motion under Gravity :

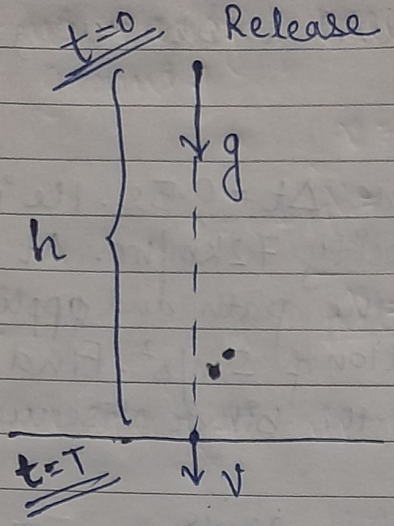


$F = \frac{GMm}{R^2} = mg$

$g = \frac{GM}{R^2} = 9.8 m/s^2 \approx 10 m/s^2$

$g \approx \pi^2$

Note: Velocity of object released is same as that of velocity of person releasing it.
If not mentioned, we assume him at rest



① T = Time of flight

$u = 0$ $\quad s = +h$

$a = +g$

$s = ut + \frac{1}{2}at^2$

$\Rightarrow h = \frac{1}{2}gT^2$

$$\boxed{T = \sqrt{\frac{2h}{g}}}$$

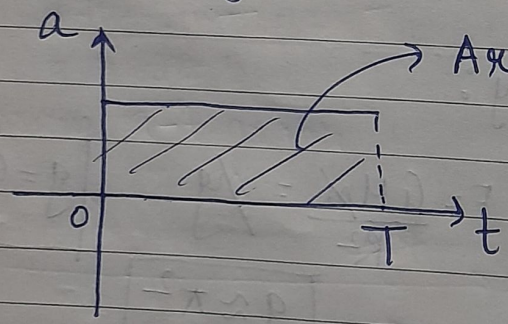
② Speed of ball when it is striking the ground.

$v = u + at$

$v = 0 + g\sqrt{\frac{2h}{g}}$

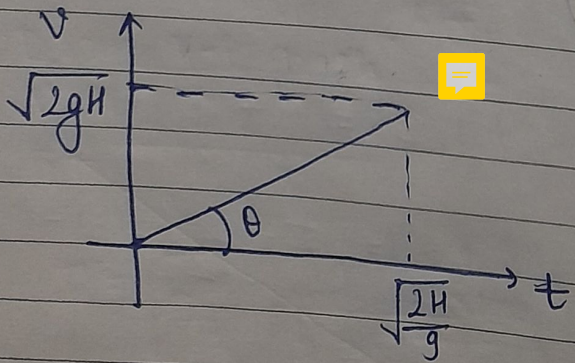
$$\boxed{v = \sqrt{2gH}}$$

③ Draw a-t graph



Area $= g\sqrt{\frac{2H}{g}} = \sqrt{2gH} = \Delta v$

④ Draw v-t graph

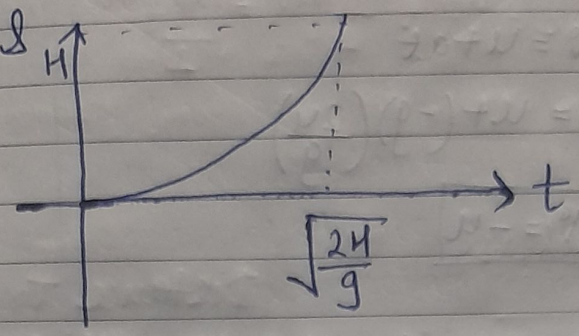


$v = 0 + gt$

$v = gt$

$\tan\theta = g$

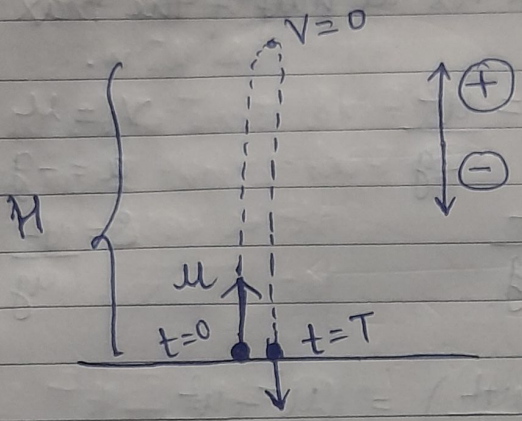
⑤ Draw s-t graph



$s = ut + \frac{1}{2}gt^2$

$s = \frac{1}{2}gt^2$

Q



① Find Maximum height

Initial velocity $= +u$

Final velocity $= 0$

$a = -g$

$s = +H$

Now, $v^2 - u^2 = 2as$

$0 = u^2 + 2(-g)H$

$$\boxed{H = \frac{u^2}{2g}}$$

② Find Time of ascent.

$v = u + at$

$0 = u - gt_A$

$$\boxed{t_A = \frac{u}{g}}$$

③ Find Time of Descent (t_D)

$$\boxed{t_A = t_D = \frac{u}{g}}$$

④ Time of flight $= t_A + t_D = \frac{2u}{g}$

Alternate

$s = 0$

u

$a = -g$

$s = ut + \frac{1}{2}at^2$

$0 = ut - \frac{1}{2}gt^2$

$= \frac{1}{2}gt^2 = ut$

$$\boxed{t = \frac{2u}{g}}$$

(5) Find velocity of ball when it is about to strike the ground.

$$v = u + at$$

$$v = u + (-g)\left(\frac{2u}{g}\right)$$

$$\boxed{v = -u}$$

(6) Find speed of ball when its at half of the max height

$$H_{max} = \frac{u^2}{2g}$$

$$\frac{H_{max}}{2} = \frac{u^2}{4g}$$

$v_i = u$

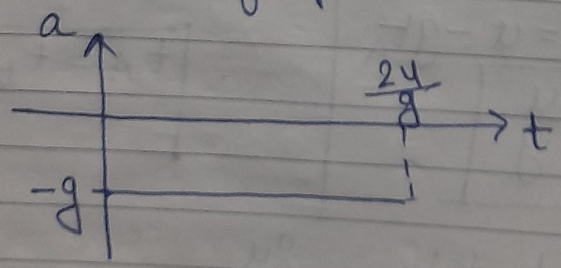
$a = -g$

$s = \frac{u^2}{4g}$

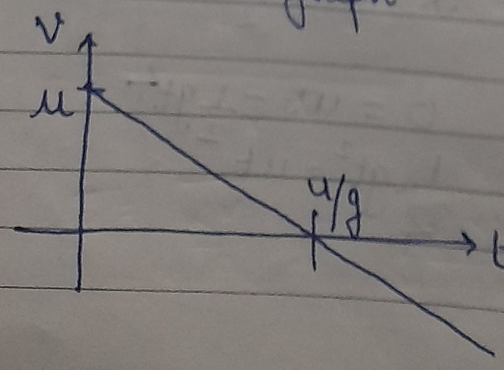
Now, $v^2 = u^2 + 2(-g)\left(\frac{u^2}{4g}\right) = u^2 - \frac{u^2}{2}$

$$v^2 = \frac{u^2}{2} \Rightarrow \boxed{v = \pm \frac{u}{\sqrt{2}}}$$

(7) Draw a-t graph



(8) Draw v-t graph



$v = u + at$

$v = u - gt$

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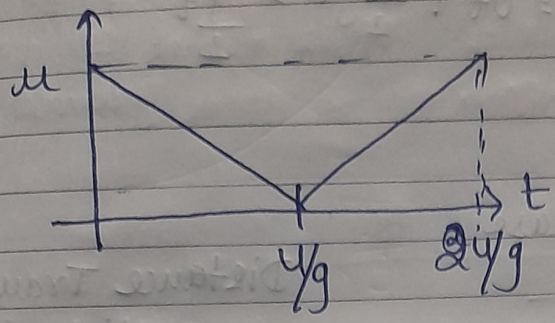
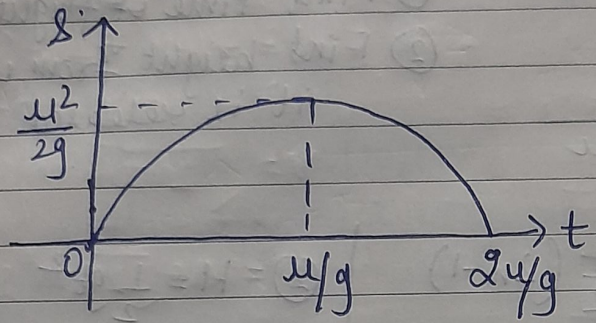
(9) Draw speed v/s t graph

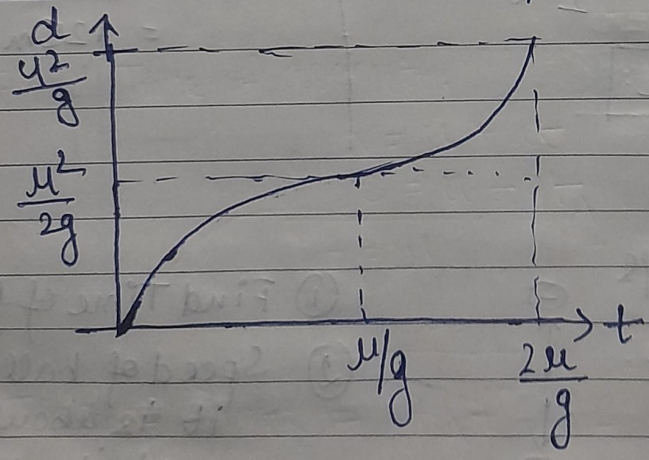


(10) Draw s-t graph

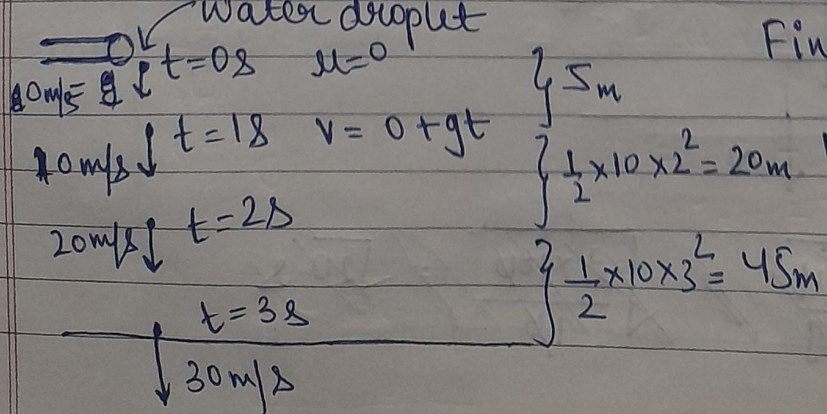
$s = ut - \frac{1}{2}gt^2$



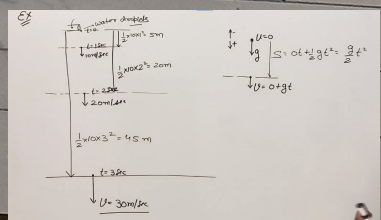
(11) Draw distance travelled v/s time



Q



Find velocity and distance travelled by each droplet at this instant.



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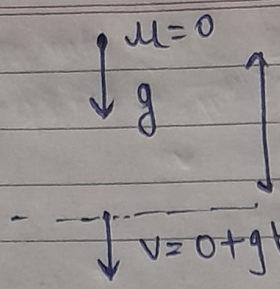
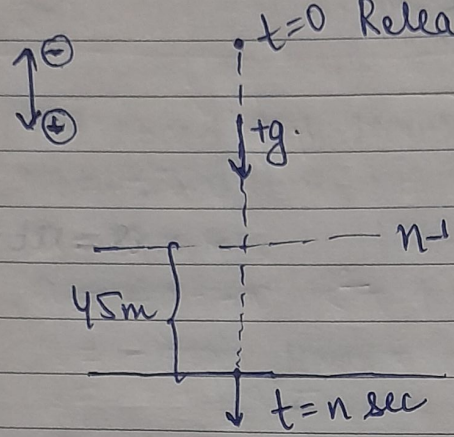


$$s = 0t + \frac{1}{2}gt^2 = \frac{g}{2}t^2$$

Q



Distance Travelled by particle in n^{th} sec is 45m.

① Find Time of flight
② Find height from which it is released.

① $S_n = u + \frac{a}{2}(2n-1)$

$= 45 = 0 + \frac{10}{2}(2n-1)$

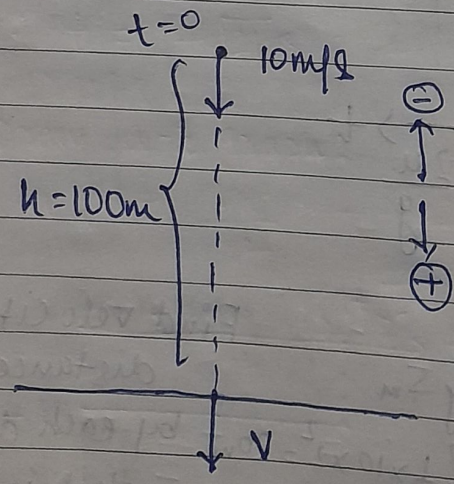
$\boxed{n = 5 \text{ sec}}$

② $H = \frac{1}{2}gt^2$

$= \frac{1}{2} \times 10 \times 5^2$

$= 125 \text{m}$

Q



① Find Time of flight
② Speed of ball when it is about to strike the ground.

① $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 100}{10}} = \sqrt{20}\ s$

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① $s = +100\,m$
$u = +10\,m/s$
$a = +g$
$T = ?$

$s = uT + \frac{1}{2}aT^2$

$\Rightarrow 100 = 10T + \frac{1}{2} \times 10 \times T^2$

$= 5t^2 + 10t - 100 = 0$

$= t^2 + 2t - 20 = 0$

$= t = \frac{-2 \pm \sqrt{4 + 4 \times 20}}{2}$

$t = -1 \pm \sqrt{21}\,s$

$\therefore t = -1 + \sqrt{21}\,s$

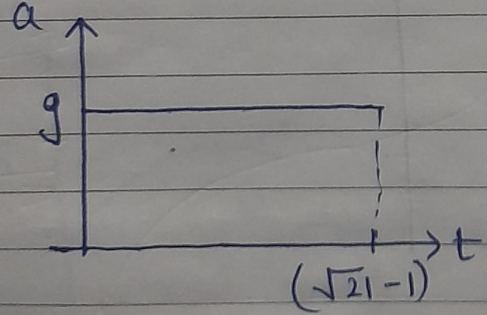
② $v^2 = u^2 + 2as$

$v^2 = 10^2 + 2(10)(100)$

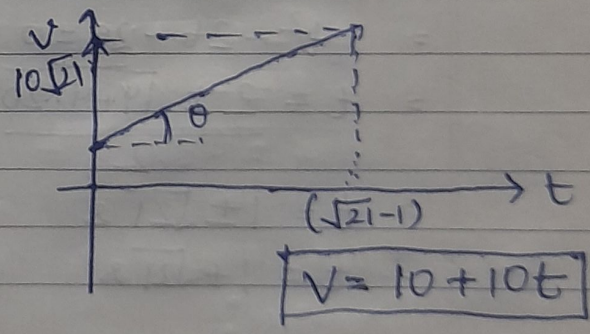
$v^2 = 100 + 2000$

$v = \sqrt{2100} = 10\sqrt{21}\,m/s$

③ Draw a-t graph:

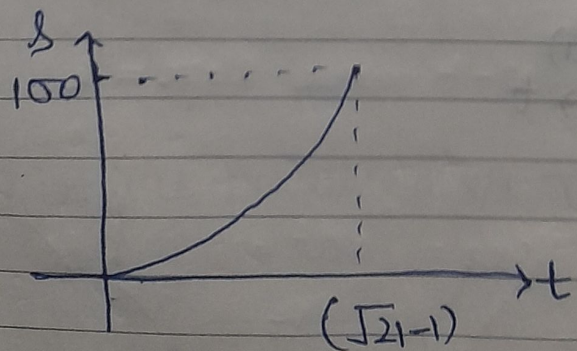


④ Draw v-t graph



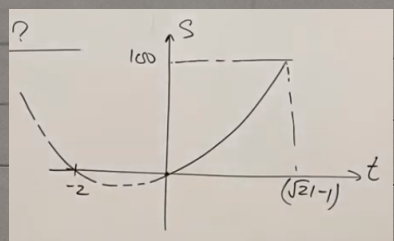
$v = 10 + 10t$

⑤ Draw s v/s t



$s = 10t + \frac{1}{2}10t^2$

$s = 10t + 5t^2$



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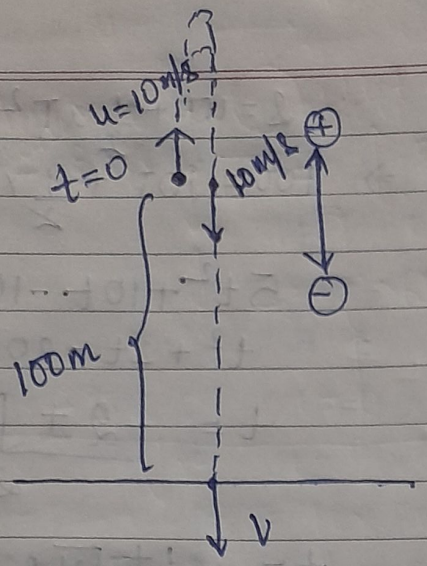


① Find Time of flight
② Find speed of ball when it is about to strike the ground

① $u = 10\ m/s$
$a = -g = -10\ m/s^2$
$s = -100\ m$

$s = ut + \frac{1}{2}gt^2$

$-100 = 10t - \frac{1}{2}(10)t^2$

$= -5t^2 + 10t + 100 = 0$

$= t^2 - 2t - 20 = 0$

$= t = \frac{2 \pm \sqrt{4 + 4 \times 20}}{2}$

$= t = 1 \pm \sqrt{21}\ s$

$\therefore t = 1 + \sqrt{21}\ s$

② $v^2 = u^2 + 2as$

$v^2 = (-10)^2 + 2(-10)(-100)$

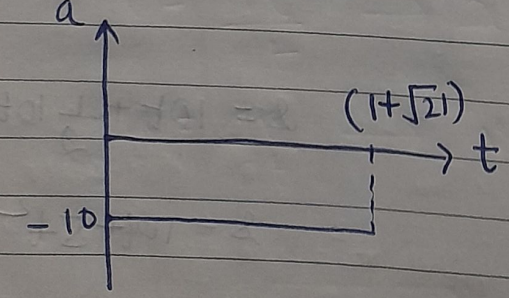
$v^2 = 100 + 2000$

$v = -\sqrt{2100}\ m/s$

$v = -10\sqrt{21}\ m/s$

-ve since the velocity is downwards

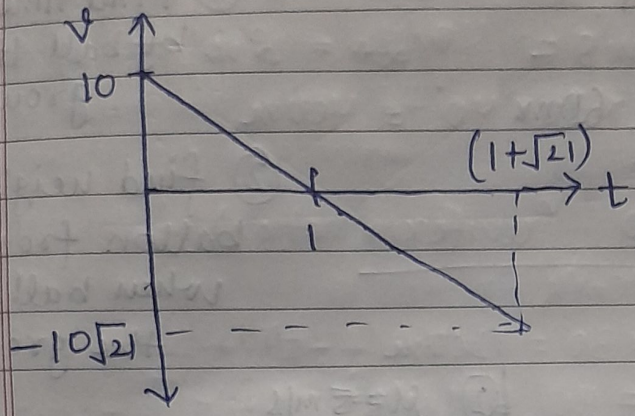
③ Draw a-t graph:



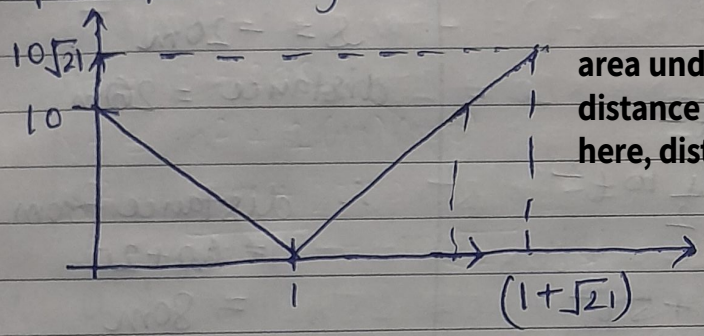
④ Draw v-t graph:

$v = u + at$

$v = 10 - 10t$



⑤ Speed v/s time graph:



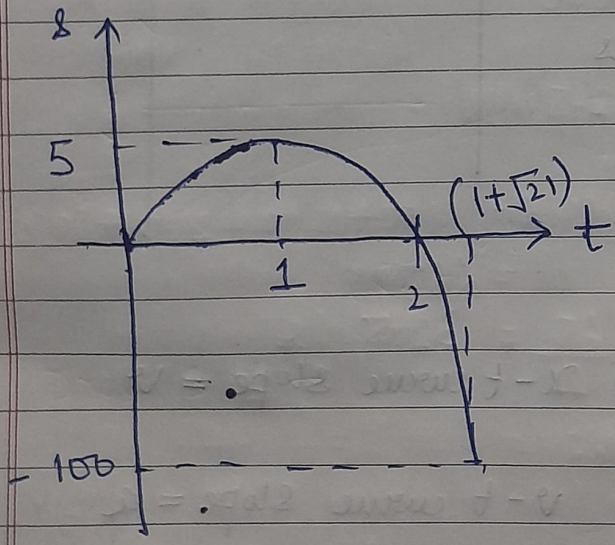
area under this curve gives the distance traveled
here, distance = 5 + 105 = 110m

⑥ Draw s-t graph

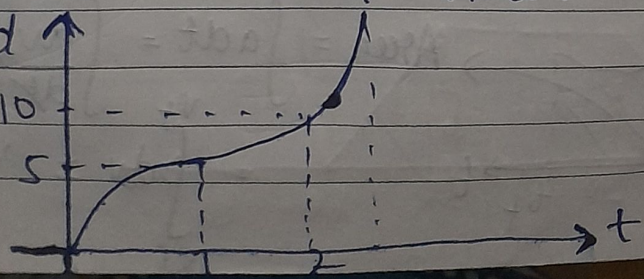
$s = ut + \frac{1}{2}gt^2$

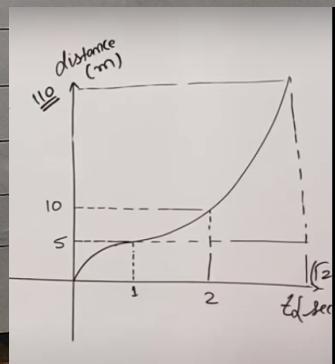
$s = 10t - \frac{1}{2}(10)t^2$

$s = 10t - 5t^2$



⑦ Distance travelled v/s Time:





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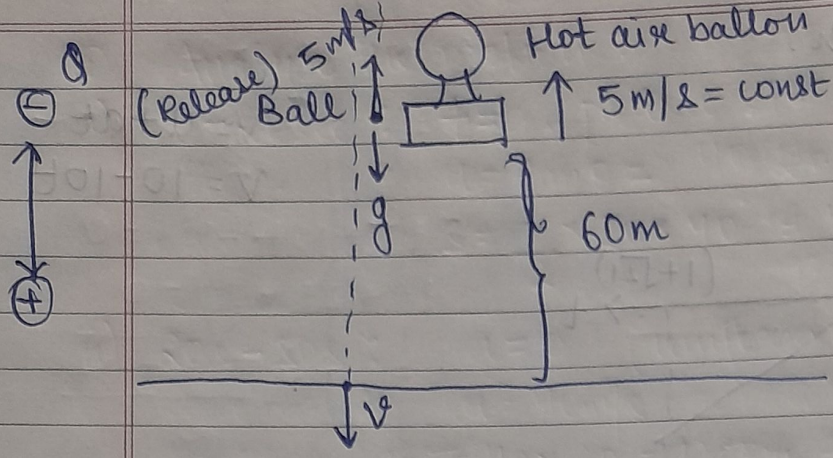


① Find time taken by ball to reach ground

② find height of ballon from ground when ball strikes the ground.

① $u = -5\,m/s$
$g = +10\,m/s$
$s = +60m$

$s = ut + \frac{1}{2}at^2$
$60 = -5t + \frac{1}{2}10t^2$

$= 60 = -5t + 5t^2$
$= 5t^2 - 5t - 60 = 0$
$= t^2 - t - 12 = 0$
$= t = -3$ or $t = 4s$
$= \boxed{t = 4sec}$

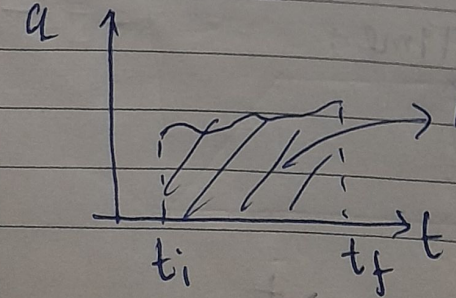
② $u = 5\,m/s$
$t = 4s$
$s = -20m$
distance $= 20m$

∴ distance from ground
$= 60 + 20$
$= 80m$

Graphs:

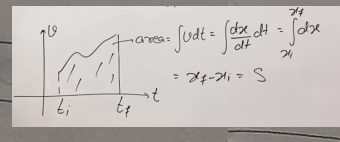
$v = \frac{dx}{dt} \longrightarrow$ x-t curve slope $= v$

$a = \frac{dv}{dt} \longrightarrow$ v-t curve slope $= a$

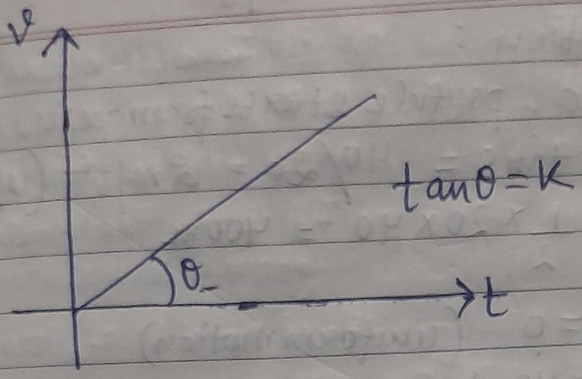


$\text{Area} = \int a\,dt = \int \frac{dv}{dt} \times dt$

$= \int_{v_i}^{v_f} dv = v_f - v_i$



Q



Comment on Acc^n

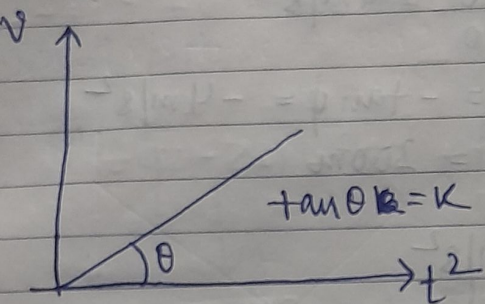
$y = mx + c^{\nearrow 0}$

$\Rightarrow v = kt$

$a = \frac{dv}{dt} = K = \text{const}$

$\therefore a = \text{const}$

Q



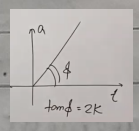
Comment on Acc^n

$y = mx + c^{\nearrow 0}$

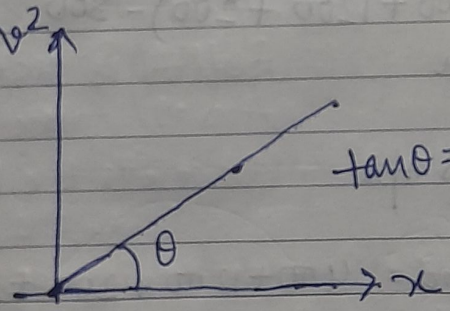
$\Rightarrow v = kt^2$

$a = \frac{dv}{dt} = 2kt$

$\therefore$ a is changing linearly.



Q



Comment on Acc^n

$y = mx + c^{\nearrow 0}$

$v^2 = kx$

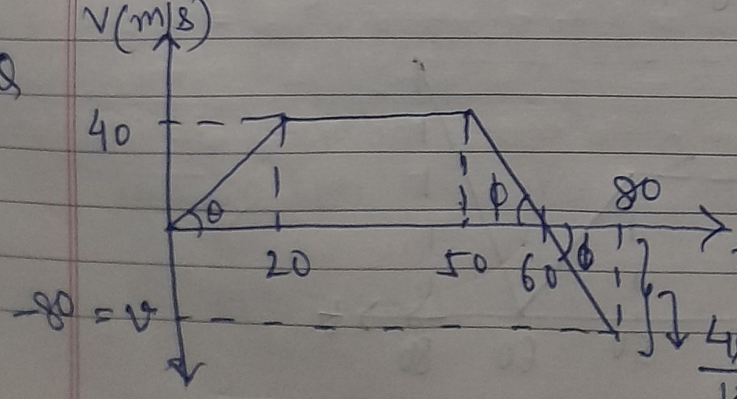
Diff wrt x

$a = \frac{2v\,dv}{dx} = k$

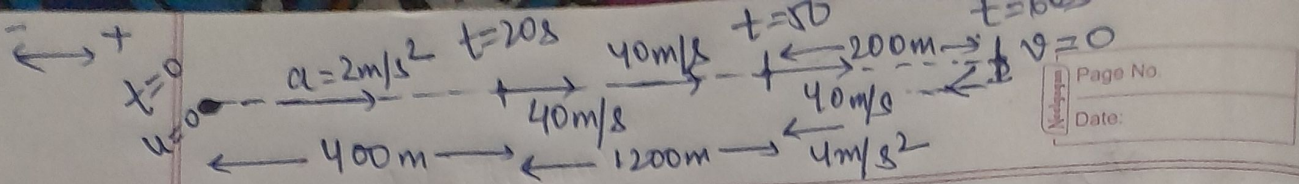
$\therefore \boxed{a = \frac{k}{2}}$

$\therefore a = \text{const}$

Q



A particle moving along straight line has vel. as given in graph.



① Analyse the motion:

i) 0-20sec: $t=0$, particle starts from rest

$a = \tan\theta = 40/20 = 2 m/s^2$ (speeding up)

$s = \frac{1}{2} \times 20 \times 40 = 400m$

ii) 20s-50s: $a = 0$ (uniform motion)

$u = 40 m/s$

$s = 40 \times 30 = 1200m$

iii) 50s-60s: $\tan\phi = \frac{40}{10} = 4 m/s^2$

$a = \tan(\pi - \phi) = -\tan\phi = -4 m/s^2$

$s = \frac{1}{2} \times 10 \times 80 = 200m$

a=+ve v=+ve speed up
a=-ve v=-ve speed up
a=+ve v=-ve speed down
a=-ve v=+ve speed down

iv) 60s-80s: $a = -4 m/s^2$

$s = \frac{1}{2} \times 20 \times -80 = -800m$

② Total displacement ⇒ $(400 + 1200 + 200) - 800$

$s = 1000m$

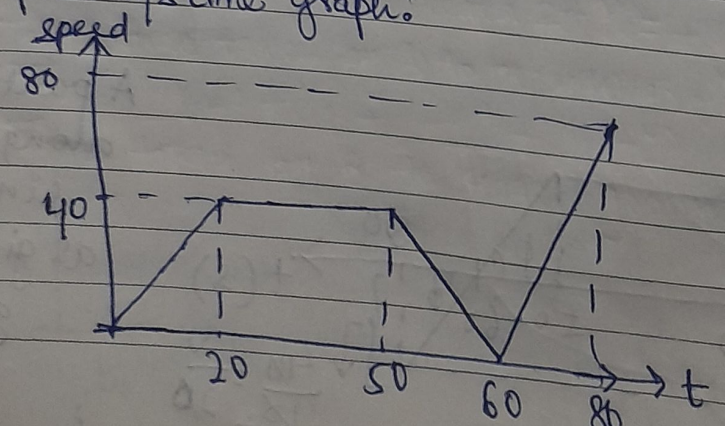
③ $V_{avg} = \frac{1000}{80} = 12.5 m/s$

④ Total distance travelled = $400 + 1200 + 200 + 800$

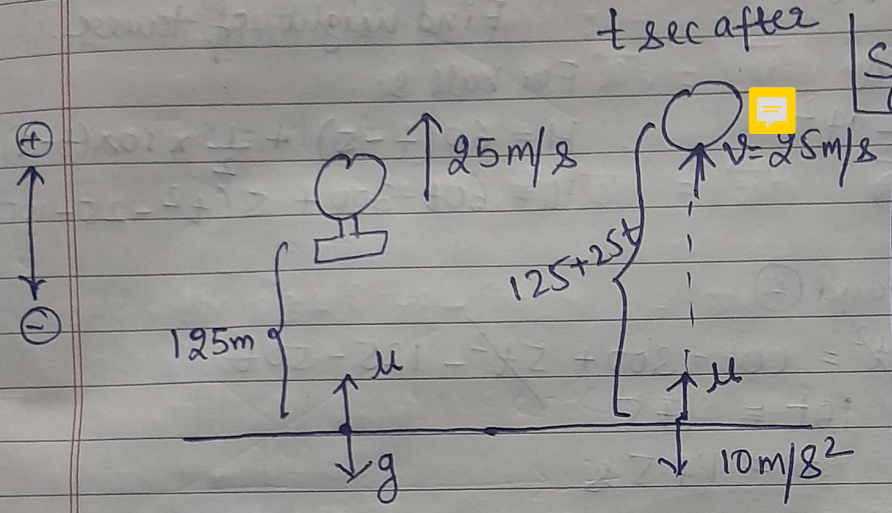
$d = 2600m$

⑤ Avg speed = $\frac{2600}{80} = \frac{65}{2} = 32.5 m/s$

⑥ Speed v/s time graph:



Q A hot air ballon starts moving vertically upwards with constant velocity 25 m/s. After 5 sec a bullet is shot vertically upwards from ground. Find minm velocity of bullet at which it can hit the ballon.



$S = 25 \times 5 = 125\text{m}$ (Ballon)

$v = u + at$

$25 = u - gt$

$\boxed{25 = u - 10t}$ — ①

$t = \frac{u-25}{10}$

$s = ut + \frac{1}{2}at^2$

$= 125 + 25t = ut - \frac{1}{2}10t^2$ — ②

$= 125 + 25\left(\frac{u-25}{10}\right) = u\left(\frac{u-25}{10}\right) - \frac{1}{2}(10)\left(\frac{u-25}{10}\right)^2$

$= 125 + \frac{25u}{10} - \frac{125}{2} = \frac{u^2}{10} - \frac{5}{2}u - 5 \left\{\frac{u^2+625-50u}{100}\right\}$

$= \frac{125}{2} + \frac{5}{2}u = \frac{u^2}{10} - \frac{5u}{2} - \frac{u^2}{20} - \frac{625}{20} + \frac{50u}{20}$

$= \frac{u^2}{20} - \frac{5u}{2} - \frac{625}{20} - \frac{125}{2} = 0$

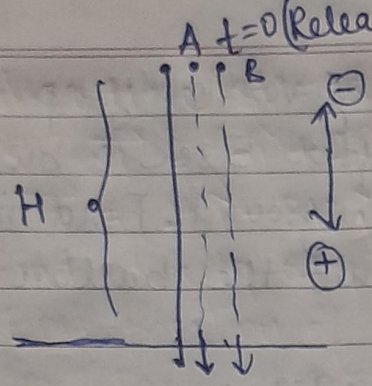
$= u^2 - 50u - 1875 = 0$

$= u = \frac{50 \pm \sqrt{2500 + 4 \times 1875}}{2} = \frac{50 \pm \sqrt{2500 + 7500}}{2}$

$= u = \frac{50 \pm 100}{2} \Rightarrow u = 75, -25$

$\therefore u = 75\ \text{m/s}$

Q



at $t = 5$ sec, second ball B is thrown vertically down with initial velocity 60 m/s. If both the balls reach ground simultaneously. Find height of tower.

For ball A

$H = 0t + \frac{1}{2}gt^2$

$H = 5t^2$ — ①

For ball B

$H = 60(t-5) + \frac{1}{2} \times 10 \times (t-5)^2$

$H = 60t - 300 + 5(t^2 - 25 - 10t)$ — ②

From ① and ②

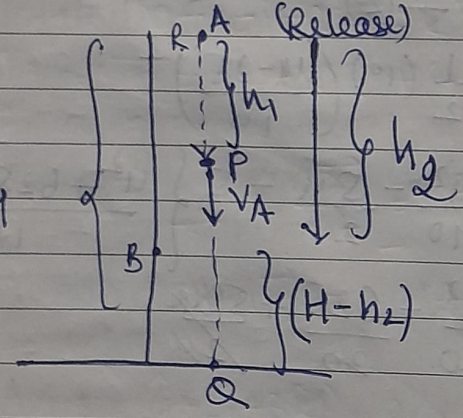
$5t^2 = 60t - 300 + 5t^2 - 125 - 50t$

$10t = 175$

$t = 17.5$ s

$\therefore H = 5 \times (17.5)^2 = 1531.25$ m

Q



Ball A is released from top of tower. When it descends by h_1, ball B is released from h_2 below the top of tower. If both balls reach ground simult.. Find height of tower

Time taken for A to reach pt P = $\sqrt{\frac{2h_1}{g}}$

Total time taken by A to reach ground = $\sqrt{\frac{2H}{g}}$

Time taken by B to reach ground = $\sqrt{\frac{2(H-h_2)}{g}}$

time taken by A to reach from pt. P to ground = time taken by B to reach ground

$$\Rightarrow \sqrt{\frac{2(H-h_2)}{g}} = \sqrt{\frac{2H}{g}} - \sqrt{\frac{2h_1}{g}}$$

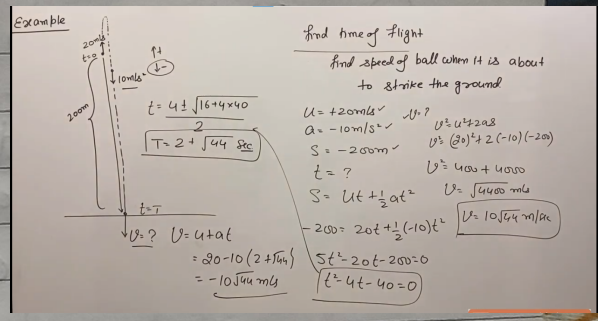
$$= \quad H - h_2 = H + h_1 - 2\sqrt{Hh_1}$$

$$= \quad 2\sqrt{Hh_1} = h_1 + h_2$$

$$= \quad 4Hh_1 = (h_1+h_2)^2$$

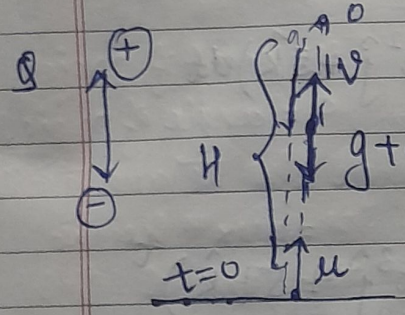
$$\therefore \boxed{H = \frac{(h_1+h_2)^2}{4h_1}}$$

Example



find time of flight

find speed of ball when it is about to strike the ground



Due to force of air resistance there is extra acceleration 'a' of ball in air against its velocity.

Find ① Time of ascent
② Time of descent
③ Max^m Height

① $v = 0$, u, $a = -(g+a)$

$v = u + at_A$

$0 = u - (g+a)t_A$

$$\boxed{t_A = \frac{u}{g+a}}$$

② $t_{Descent}$

$a = -(g-a)$

$u = 0$

$s = -H = \frac{-u^2}{2(g+a)}$

$$\frac{-u^2}{2(g+a)} = 0 - \frac{1}{2}(g-a)t_D^2$$

$$\boxed{t_D = \frac{u}{\sqrt{g^2-a^2}}}$$

③ $s = +H$

~~$s = ut + \frac{1}{2}gt^2$~~

$v^2 = u^2 + 2as$

$0 = u^2 - 2(g+a)H$

$$\boxed{H = \frac{u^2}{2(g+a)}}$$

$$\frac{t_a}{t_d} = \frac{u\sqrt{(g-a)(g+a)}}{(g+a)u} = \sqrt{\frac{g-a}{g+a}} = \frac{t_a}{t_d}$$

④ Find speed of ball when it is about to strike the ground

$v^2 = 0 + 2(g-a)\frac{u^2}{2(g+a)}$

$$\boxed{v = u\sqrt{\frac{g-a}{g+a}}}$$

Q A particle starts from $(1,2,3)$ m with initial velocity $\vec{u} = (-\hat{i}+2\hat{j}+3\hat{k})$ m/s having variable $\vec{a} = (2t\hat{i}-3t^2\hat{j}+5\hat{k})$ m/s^2 at any time t.

① Find velocity of particle

② Find position of particle

①
$$\frac{d\vec{v}}{dt} = \vec{a}$$
$$\int_{(-\hat{i}+2\hat{j}+3\hat{k})}^{\vec{v}} d\vec{v} = \int_0^t \vec{a}\,dt$$
$$= \int_{(-\hat{i}+2\hat{j}+3\hat{k})}^{\vec{v}} d\vec{v} = \int_0^t (2t\hat{i}-3t^2\hat{j}+5\hat{k})\,dt$$
$$= (\vec{v})_{(-\hat{i}+2\hat{j}+3\hat{k})}^{\vec{v}} = (t^2\hat{i}-t^3\hat{j}+5t\hat{k})_0^t$$
$$= \vec{v}-(-\hat{i}+2\hat{j}+3\hat{k}) = t^2\hat{i}-t^3\hat{j}+5t\hat{k}$$
$$= \vec{v} = (t^2-1)\hat{i}+(-t^3+2)\hat{j}+(5t+3)\hat{k}$$

②
$$\frac{d\vec{r}}{dt} = (t^2-1)\hat{i}+(-t^3+2)\hat{j}+(5t+3)\hat{k}$$
$$\int_{(\hat{i}+2\hat{j}+3\hat{k})}^{\vec{r}} d\vec{r} = \int_0^t \left((t^2-1)\hat{i}+(-t^3+2)\hat{j}+(5t+3)\hat{k}\right)dt$$
$$= \vec{r}-(\hat{i}+2\hat{j}+3\hat{k}) = \left(\frac{t^3}{3}-t\right)\hat{i}+\left(\frac{-t^4}{4}+2t\right)\hat{j}+\left(\frac{5t^2}{2}+3t\right)\hat{k}$$
$$\vec{r} = \left(\frac{t^3}{3}-t+1\right)\hat{i}+\left(\frac{-t^4}{4}+2t+2\right)\hat{j}+\left(\frac{5t^2}{2}+3t+3\right)\hat{k}$$

Q A particle moving along x-axis starts from rest from point $x = \pm A$. It has variable acceleration which varies with its position as $a = -\omega^2 x$. Find its velocity as f^n of x, velocity at any time t and position at any time t.

$$a = -\omega^2 x$$

$$v\frac{dv}{dx} = -\omega^2 x$$

$$\int_0^v v\,dv = -\omega^2 \int_{+A}^{x} x\,dx$$

$$\frac{v^2}{2} = \frac{-\omega^2}{2}(x^2 - A^2)$$

$$\boxed{v = \omega\sqrt{A^2 - x^2}}$$

$v = \omega\sqrt{A^2 - x^2}$

$x = 0 \Rightarrow v_{max} = \omega A$

$v = 0 \Rightarrow x^2 = A^2$

$x = \pm A$

$$v = \frac{dx}{dt} = \omega\sqrt{A^2 - x^2}$$

$$\int_A^x \frac{dx}{\sqrt{A^2 - x^2}} = \omega \int_0^t dt$$

$$= \left[\sin^{-1}\left(\frac{x}{A}\right)\right]_A^x = \omega (t)_0^t$$

$$\int \frac{dx}{\sqrt{A^2 - x^2}} \qquad x = A\sin\theta, \quad dx = A\cos\theta\, d\theta$$

$$\int \frac{A\cos\theta\, d\theta}{\sqrt{A^2 - A^2\sin^2\theta}}$$

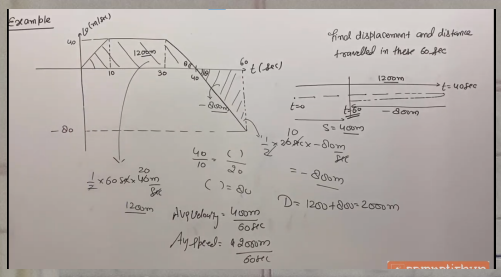
$$\int \frac{A\cos\theta\, d\theta}{A\cos\theta} = \theta = \sin^{-1}\frac{x}{A}$$

$$= \sin^{-1}\left(\frac{x}{A}\right) - \sin^{-1}(1) = \omega t$$

$$= \sin^{-1}\left(\frac{x}{A}\right) - \frac{\pi}{2} = \omega t$$

$$= \sin^{-1}\left(\frac{x}{A}\right) = \omega t + \frac{\pi}{2}$$

$$= \boxed{x = A\sin\left(\omega t + \frac{\pi}{2}\right) = A\cos\omega t}$$

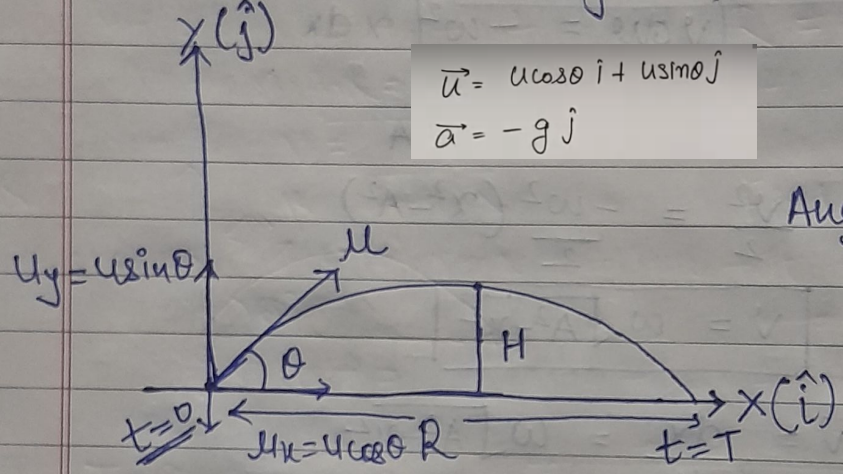


$$v = \frac{dx}{dt} = -A\omega \sin \omega t$$

Projectile Motion

Galileo's statement : Motion in mutually ⊥ direcn are independent of each other.

Ground to Ground Projectile Motion :



$\vec{u} = u\cos\theta\, \hat{i} + u\sin\theta\, \hat{j}$

$\vec{a} = -g\hat{j}$

u = speed of projection

θ = angle of projection

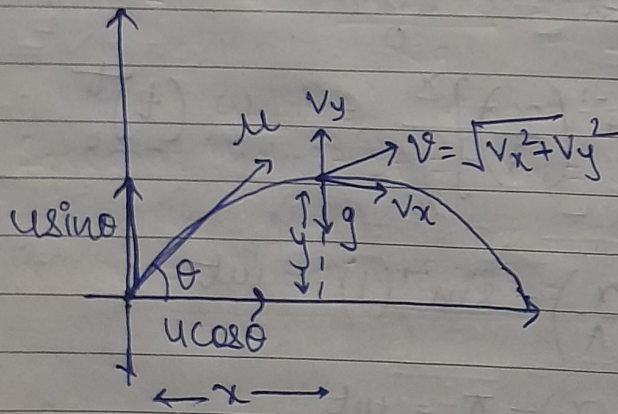
Angle b/w $\vec{u}$ and $\vec{g} = 90 + \theta$

H = max^m height

T = Time of flight

R = Range

Finding parameters in projectile motion at any time 't' :



x	y
$u_x = u\cos\theta$	$u_y = u\sin\theta$
$a_x = 0$	$a_y = -g$
$v_x = u_x + g_x t$	$v_y = u_y + a_y t$
$v_x = u\cos\theta$	$v_y = u\sin\theta - gt$
$v_x^2 = u_x^2 + 2g_x x$	$v_y^2 = u_y^2 + 2a_y y$
$v_x = u_x = u\cos\theta$	$v_y^2 = u^2\sin^2\theta - 2gy$
$x = ut + \frac{1}{2}g_x t^2$	$y = u_y t + \frac{1}{2}a_y t^2$
$x = u\cos\theta\, t$	$y = (u\sin\theta)t - \frac{1}{2}gt^2$

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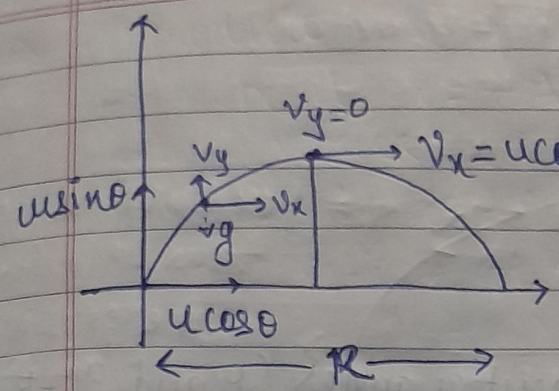


Time of ascent (t_A):

$v_y = 0 \quad u_y = u\sin\theta$

$a_y = -g$

$v_y = u_y + a_y t_A$

$0 = u\sin\theta - g t_A$

$$\boxed{t_A = \frac{u\sin\theta}{g}} \qquad \boxed{t_A = \frac{u_y}{g}}$$

Time of descent (t_d)

$t_d = t_A = \frac{u\sin\theta}{g}$

$$\therefore \boxed{\text{Time of flight } (T) = \frac{2u\sin\theta}{g} = \frac{2u_y}{g}}$$

$y = u_y t + \frac{1}{2} a_y t^2$

$y = (u\sin\theta)t - \frac{1}{2} g t^2 = 0$

$$\boxed{t = \frac{2u\sin\theta}{g} = \frac{2u_y}{g}}$$

Maximum Height (H):

$u_y = u\sin\theta$

$v_y = 0$

$a_y = -g$

$y = H$

$v_y^2 - u_y^2 = 2 a_y y$

$0 - (u\sin\theta)^2 = 2(-g) H$

$$\boxed{H = \frac{(u\sin\theta)^2}{2g} = \frac{u_y^2}{2g}}$$

$y = u_y t + \frac{1}{2} a_y t^2$

$y = (u\sin\theta)t - \frac{1}{2} g t^2 = u\sin\theta\left(\frac{u\sin\theta}{g}\right) - \frac{1}{2}\left(\frac{u\sin\theta}{g}\right)^2$

y_{max}

$\frac{dy}{dt} = 0 = u\sin\theta - \frac{1}{2} g (2t)$

$t = \frac{u\sin\theta}{g}$

$$\boxed{Y_{max} = \frac{u^2\sin^2\theta}{2g} = H}$$

Range (R):

$R = u_x t + \frac{1}{2} a_x t^2$ ($a_x = 0$)

$R = u_x T$

$R = u_x \left(\frac{2u_y}{g}\right)$

$$\boxed{R = \frac{2(u\cos\theta)(u\sin\theta)}{g} = \frac{u^2\sin 2\theta}{g}}$$

$$\boxed{R = \frac{2u_x u_y}{g}}$$

Note:

⇒ $R_{max} \Rightarrow 2\theta = 90^\circ$

$\theta = 45^\circ$

$$\boxed{R_{max} = \frac{u^2}{g}}$$

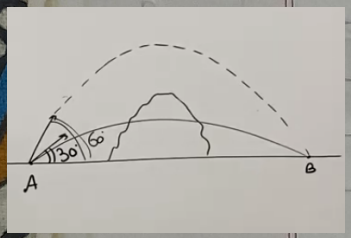
⇒ Range is same at complimentry angles (θ and $90-\theta$)

$$R = \frac{2u^2 \sin\theta\cos\theta}{g}$$

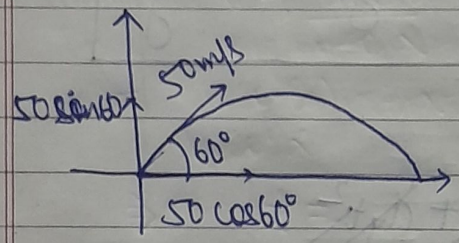
$$R' = \frac{2u^2 \sin(90-\theta)\cos(90-\theta)}{g}$$

$$R' = \frac{2u^2 \cos\theta \sin\theta}{g}$$

$$\therefore \boxed{R = R'}$$



Q



Find ① Time of flight
② Hmax
③ Speed at Hmax
④ Range

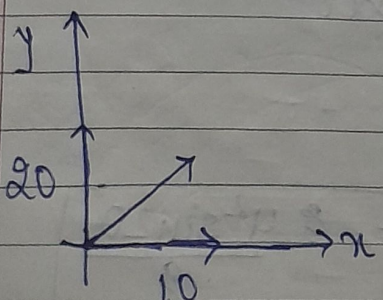
① $T = \frac{2u\sin\theta}{g} = \frac{2 \times 50 \times \sqrt{3}/2}{10} = 5\sqrt{3}\ s$

② $H_{max} = \frac{u^2\sin^2\theta}{2g} = \frac{\overset{25}{\cancel{50}} \times 50 \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}}{2 \times 10} = \frac{375}{4}\ m$

③ Speed at Highest pt $= V_x = u\cos\theta = 50 \times \frac{1}{2} = 25\ m/s$

④ $Range = \frac{2u^2\sin\theta\cos\theta}{g} = \frac{\overset{25}{\cancel{2}} \times \cancel{50} \times 50 \times \frac{\sqrt{3}}{\cancel{2}} \times \frac{1}{\cancel{2}}}{10} = 125\sqrt{3}\ m$

Q



A ball is projected with initial velocity $\vec{u} = (10\hat{i} + 20\hat{j})$ m/s

Find ① T ② Hmax
③ Speed at Hmax ④ Range

$\vec{u} = 10\hat{i} + 20\hat{j}$

$|\vec{u}| = \sqrt{100+400} = \sqrt{500} = 10\sqrt{5}$ m/s

$\tan\theta = \frac{20}{10} = 2$

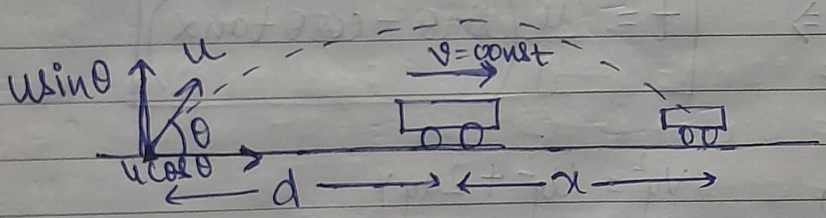
① $T = \frac{2u_y}{g} = \frac{2\times 20}{10} = 4s$

② $H_{max} = \frac{(u_y)^2}{2g} = \frac{20\times 20}{20} = 20m$

③ $V_{Hmax} = u_x = 10$ m/s

④ $R = \frac{2u_y u_x}{g} = \frac{2\times 10\times 20}{10} = 40m$

Q



Find the velocity with which the ball should be thrown (u)!

Neglect Height of car

Time taken by ball to land on car = $T = \frac{2u\sin\theta}{g}$

Till then the car will displace by x units.

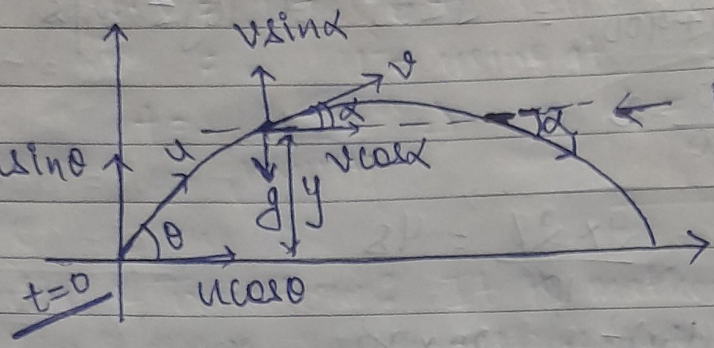
$x = vT = \frac{v\cdot 2u\sin\theta}{g}$

Range of ball = $d + x$

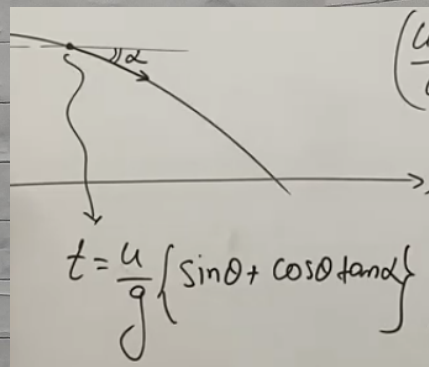
$\frac{2u^2\sin\theta\cos\theta}{g} = d + \frac{2vu\sin\theta}{g}$

solve quadratic and find value of u

How velocity varies in projectile motion?



← For this time replace tanα by tan(-α) in expression of t.



$$u\cos\theta = v\cos\alpha \quad \text{—①}$$

Now, $v_y = u_y + a_y t$

$$v\sin\alpha = u\sin\theta - gt \quad \text{—②}$$

From ① $v = \dfrac{u\cos\theta}{\cos\alpha}$

$$= \left(\frac{u\cos\theta}{\cos\alpha}\right)\sin\alpha = u\sin\theta - gt$$

$$\Rightarrow \boxed{t = \frac{u}{g}\left(\sin\theta - \cos\theta\tan\alpha\right)}$$

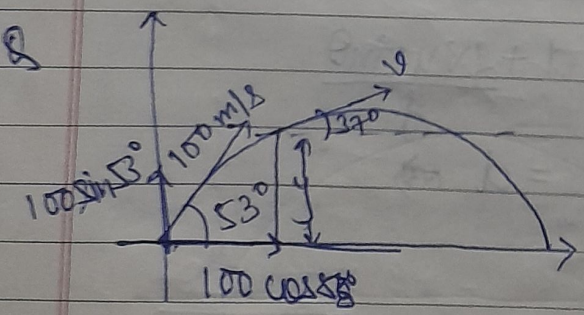
Now, $v_y^2 = u_y^2 + 2a_y y$

$$\Rightarrow v^2\sin^2\alpha = u^2\sin^2\theta - 2gy$$

$$= \left(\frac{u^2\cos^2\theta}{\cos^2\alpha}\right)\sin^2\alpha = u^2\sin^2\theta - 2gy$$

$$\boxed{y = \frac{u^2}{2g}\left(\sin^2\theta - \cos^2\theta\tan^2\alpha\right)}$$

Q



When ball is moving at an angle 37° with horizontal Find its speed, time and height.

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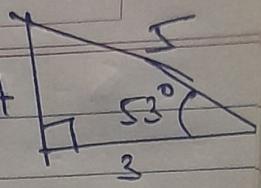


① speed: $u_x = v_x$

$u\cos\theta = v\cos\alpha$

$100\cos 53 = v\cos 37°$

$\frac{100\cos 53}{\cos 37°} = v \Rightarrow v = \frac{100\cos 53}{\sin 53}$

$v = 100\cot 53° = \frac{100 \times 3}{4} = \frac{300}{4} = 75\ m/s$

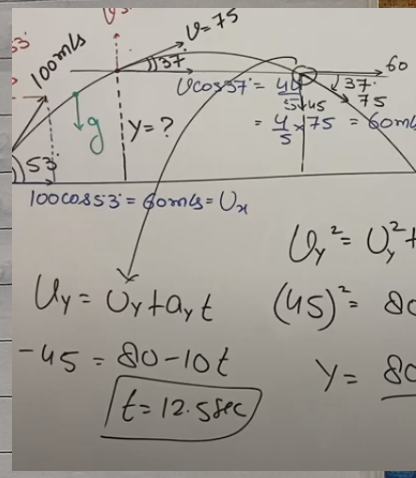
② time: $v_y = u_y + a_y t$

$75\sin 37° = 100\sin 53° - gt$

$\frac{75 \times 3}{5} = \frac{100 \times 4}{5} - 10t$

$45 = 80 - 10t$

$t = \frac{35}{10} = 3.5\ s$

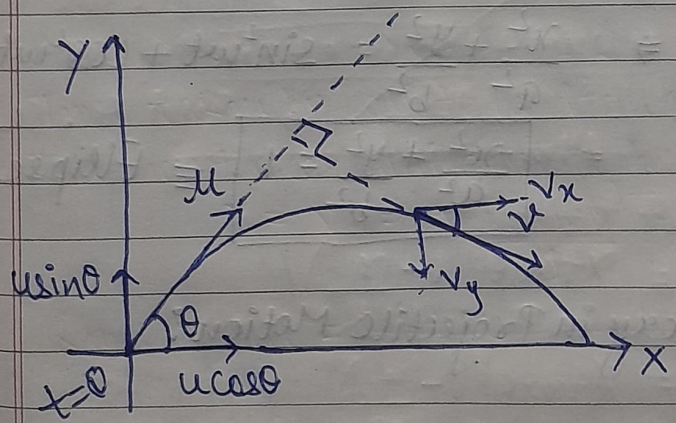


③ Height = $v_y^2 = u_y^2 - 2gs$

$= (45)^2 = (80)^2 - 2(10)y$

$y = \frac{(80)^2 - (45)^2}{20} = \frac{125 \times 35}{20 \times 4} = \frac{875}{4}\ m$

Q



Find the time after which particle will be moving ⊥ to initial direction of motion.

izontal

$\vec{u} = u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}$

We know $v_x = u_x$

$\therefore v_x = u\cos\theta\,\hat{i}$

$v_y = u_y - gt$

$= u\sin\theta - gt$

$\Rightarrow \vec{v} = (u\cos\theta)\hat{i} + (u\sin\theta - gt)\hat{j}$

As $\vec{u}$ and $\vec{v}$ are $\perp \Rightarrow \vec{u}\cdot\vec{v} = 0$

$\vec{u}\cdot\vec{v} = u^2\cos^2\theta + u^2\sin\theta - (u\sin\theta)gt = 0$

$= u^2 = u\sin\theta\, gt$

$t = \frac{u^2}{(u\sin\theta)g} \Rightarrow$ $\boxed{t = \frac{u\,\mathrm{cosec}\,\theta}{g}}$

path

Equation of Trajectory: It is mathematical relation b/w instantaneous coordinates (x,y)

Ex $x = a\sin\omega t$ —① Find eqn of trajectory

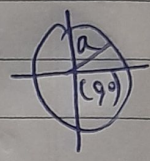
$y = a\cos\omega t$ —②

Square both side and add to eliminate t

$x^2 + y^2 = a^2(\sin^2\omega t + \cos^2\omega t)$

$x^2 + y^2 = a^2$ ≡ Circle with centre (0,0)

$r = a$



Ex $x = a\sin\omega t$ —① Find eqn of trajectory

$y = b\cos\omega t$ —②

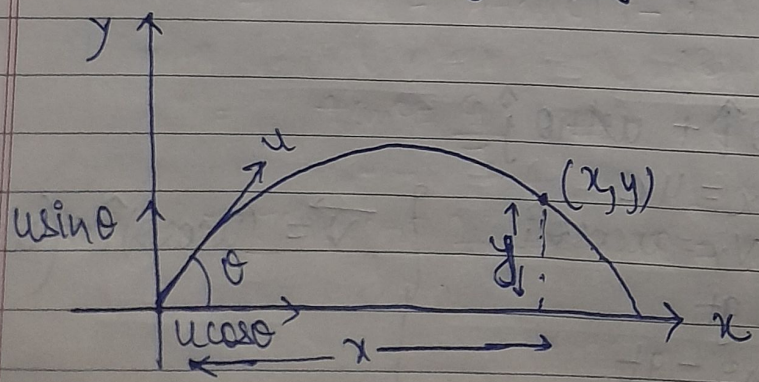
$\Rightarrow \frac{x}{a} = \sin\omega t$

$\frac{y}{b} = \cos\omega t$

$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = \sin^2\omega t + \cos^2\omega t$

$= \boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1}$ ≡ Ellipse

Equation of Trajectory in Projectile Motion:



$x = (u\cos\theta)t$ —①

$y = u_y t + \frac{1}{2}a_y t^2$

$y = (u\sin\theta)t - \frac{1}{2}gt^2$ —②

From ①: $t = \frac{x}{u\cos\theta}$

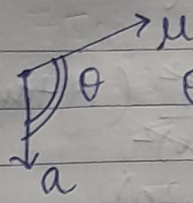
$$y = \cancel{u}\sin\theta \cdot \frac{x}{\cancel{u}\cos\theta} - \frac{1}{2} g \frac{x^2}{u^2\cos^2\theta}$$

$$\boxed{y = x\tan\theta - \frac{g x^2}{2u^2\cos^2\theta}}$$

Note: If $\vec{a}$ is constant then motion is straight line or parabola

(1) $\vec{a}$ $\rightarrow$ $\vec{v}$ $\qquad$ $\vec{a}$ $\leftarrow$ $\vec{v}$ $\Rightarrow$ Straight line

(2) otherwise



$\theta \neq 0^\circ$ or $\theta \neq 180^\circ$ $\Rightarrow$ Parabola

Now, $\quad y = x\tan\theta - g\frac{x^2}{2u^2\cos^2\theta}$

$\frac{1}{\cos^2\theta} = \sec^2\theta = 1 + \tan^2\theta$

$$\therefore \boxed{y = x\tan\theta - \frac{gx^2}{2u^2}(1+\tan^2\theta)}$$ → useful when θ needs to be find.

Also, $\quad y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

$$y = x\tan\theta\left(1 - \frac{gx}{2u^2\cos\theta\sin\theta}\right)$$

$$y = x\tan\theta\left(1 - \frac{gx}{2u^2\sin 2\theta}\right) \qquad \left(R = \frac{2u^2\sin 2\theta}{g}\right)$$

$$\therefore \boxed{y = x\tan\theta\left(1 - \frac{x}{R}\right)}$$ useful when range is given/ needs to be found

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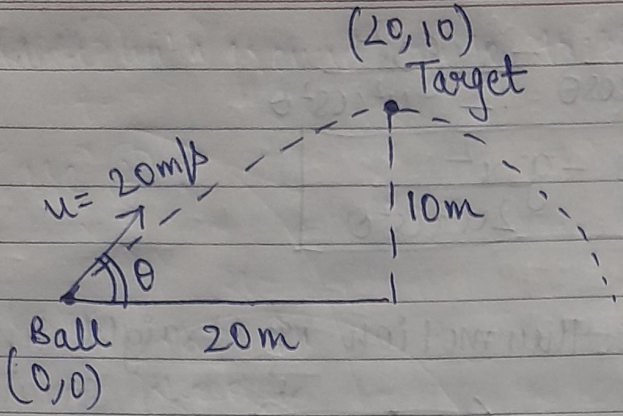
Q



Find θ so that ball strikes the target

$$y = x\tan\theta - \frac{gx^2}{2u^2}(1+\tan^2\theta)$$

$$10 = 20\tan\theta - \frac{10(20)^2}{2\times(20)^2}(1+\tan^2\theta)$$

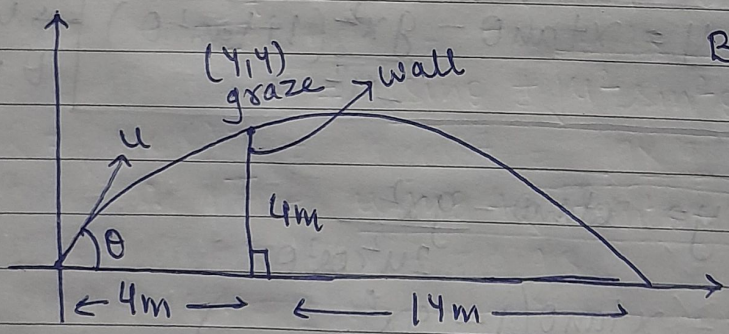
$$= \quad 2 = 4\tan\theta - (1+\tan^2\theta)$$

$$\Rightarrow \tan^2\theta - 4\tan\theta + 3 = 0$$

$$\boxed{\tan\theta = 1, 3} \Rightarrow \theta = \tan^{-1}(1) = 45^\circ$$

$$\theta = \tan^{-1}(3)$$

Q



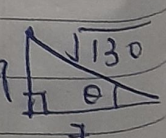
Ball just grazes top of the wall and strikes the ground 14m away from wall. Find θ, u.

R = 18m

$$y = x\tan\theta\left(1 - \frac{x}{R}\right)$$

$$\Rightarrow 4 = 4\tan\theta\left(1 - \frac{4}{18}\right)$$

$$= \tan\theta = \frac{9}{7} \Rightarrow \boxed{\theta = \tan^{-1}\left(\frac{9}{7}\right)}$$

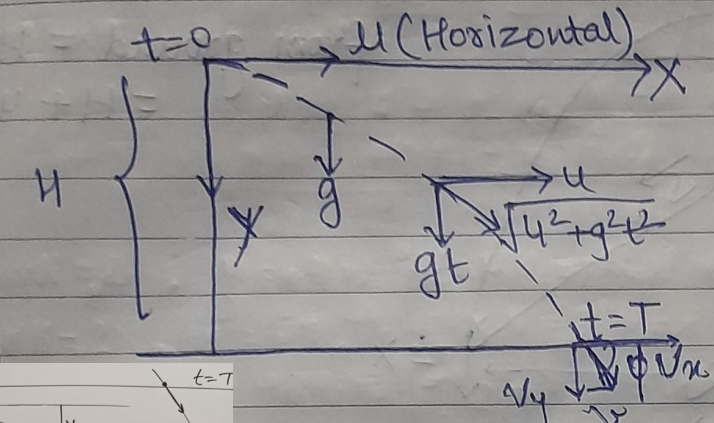


Now, $R = \frac{u^2\sin\theta\cos\theta\cdot 2}{g} \Rightarrow 18 = \frac{u^2 \times 2 \times \frac{9}{\sqrt{130}} \times \frac{7}{\sqrt{130}}}{10}$

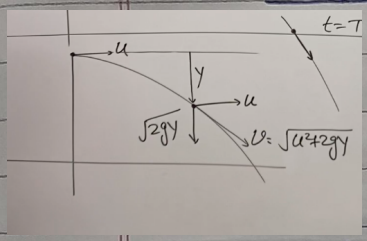
$$10 = u^2 \times \frac{7}{130} \Rightarrow u = \sqrt{\frac{1300}{7}} \text{ m/s}$$

Projection from tower:

Case ①



x	y
$u_x = u$	$u_y = 0$
$a_x = 0$	$a_y = g$
$v_x = u_x + a_x t$	$v_y = u_y + a_y t$
$v_x = u_x$	$v_y = gt$
$v_x^2 = u_x^2 + 2a_x x$	$v_y^2 = u_y^2 + 2a_y y$
$v_x = u$	$v_y = \sqrt{2gy}$
$x = u_x t + \frac{1}{2} a_x t^2$	$y = u_y t + \frac{1}{2} a_y t^2$
$x = ut$	$y = gt^2/2$



Time of flight (T):

$$y = u_y t + \frac{1}{2} a_y t^2$$

$$\Rightarrow H = 0 + \frac{1}{2} g t^2$$

$$= \boxed{T = \sqrt{\frac{2H}{g}}}$$

Horizontal Range (R):

$$R = u_x t + \frac{1}{2} a_x t \quad (a_x \to 0)$$

$$R = uT$$

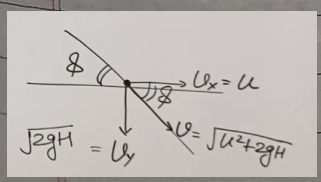
$$\boxed{R = u\sqrt{\frac{2H}{g}}}$$

Velocity of ball when it is about to strike the ground:

$$v_y^2 = u_y^2 + 2a_y y$$

$$v_y^2 = 0 + 2gH$$

$$\boxed{v_y = \sqrt{2gH}}$$



$$\boxed{v_{net} = \sqrt{u^2 + 2gH}}$$

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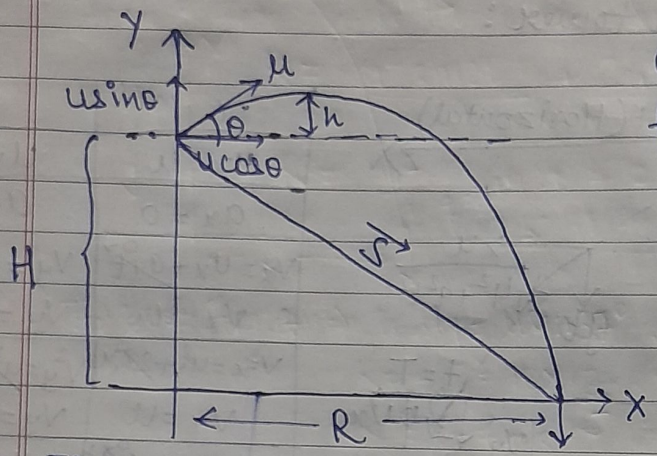
Angle with ground while striking $= \tan^{-1}\left(\frac{\sqrt{2gH}}{u}\right)$

Case (2):



(1) Maximum height from ground $= H + h$

$= H + \frac{u^2\sin^2\theta}{2g}$

(2) Time of flight (T):

$y = u_y t + \frac{1}{2} a_y t^2$

$-H = u\sin\theta\, t - \frac{1}{2} g t^2$

$\frac{g}{2}t^2 - (u\sin\theta)t - H = 0$

$t = \frac{u\sin\theta \pm \sqrt{u^2\sin^2\theta + 2gh}}{g}$ (Reject -ve)

$\therefore \boxed{T = \frac{u\sin\theta + \sqrt{u^2\sin^2\theta + 2gh}}{g}}$

$T = \frac{u\sin\theta}{g} + \sqrt{\frac{2}{g}\left(H + \frac{u^2\sin^2\theta}{2g}\right)}$

(3) Range (R): $R = u_x t + \frac{1}{2} a_x t^2$ ($a_x = 0$)

$\boxed{R = u\cos\theta\, T}$

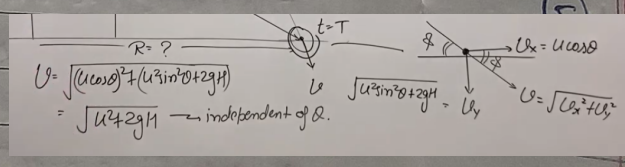
(4) Velocity of ball when it is abt to strike the ground.

$v_x = u\cos\theta$

$v_y^2 = u_y^2 + 2as$

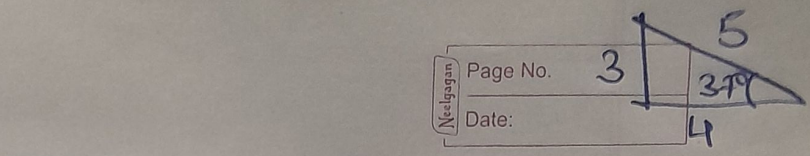
$v_y^2 = u^2\sin^2\theta + 2gH$

$v_y = \sqrt{u^2\sin^2\theta + 2gH}$

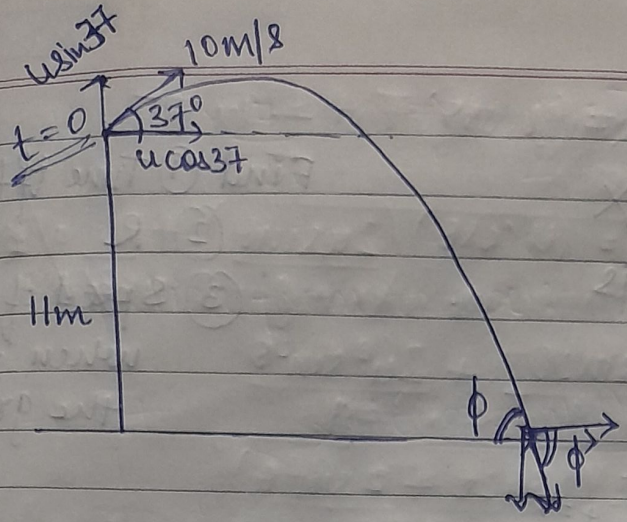


$\tan\theta = \frac{u_y}{u_x}$

$= \frac{\sqrt{u^2\sin^2\theta + 2gH}}{u\cos\theta}$



Q



Find ① Max height from ground
② Time of flight
③ Horizontal Range
④ Speed when strikes the surface
⑤ Angle with horizontal at which it strikes.

① $H_{max} = 11 + h = 11 + \frac{u^2 \sin^2\theta}{2g} = 11 + \frac{6^2}{2\times 10} = 11 + \frac{36}{2\times 5} = \frac{64}{5}$ m

② T: ~~$v_y = u_y + a_y t$~~

$$y = u_y t + \frac{1}{2} a_y t^2$$

$$-11 = 6t - \frac{1}{2}\times 10 \times t^2$$

$$5t^2 - 6t - 11 = 0$$

$$t = \frac{6 \pm \sqrt{36 + 4\times 5\times 11}}{10} = \frac{6+16}{10} = \frac{22}{10} = 2.2\,s$$

④ $V_x = u\cos\theta = 10\cos 37 = 10\times\frac{4}{5} = 8$ m/s

$V_y^2 = u_y^2 + 2gH$

$V_y^2 = 6^2 + 2\times 10\times 11 = 36 + 220 = 256$

$V_y = 16$ m/s

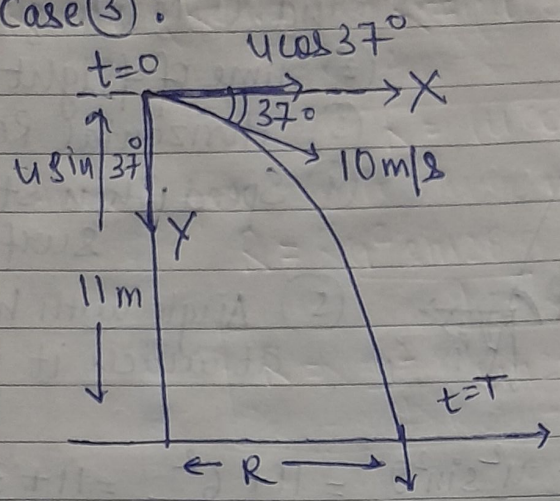
$V_{net} = \sqrt{8^2 + 16^2} = \sqrt{64+256} = \sqrt{320} = 8\sqrt{5}$ m/s

③ $R = u\cos\theta\, T = 48\times\frac{22}{10\,5} = \frac{88}{5}$ m

⑤ $\tan\phi = \frac{16}{8} \Rightarrow \phi = \tan^{-1}(2)$

Case (3):

Ex



Find (1) Time of flight
(2) R
(3) Speed of ball when it hits the ground.

(1) Time of flight ⇒ $u_y = u\sin 37°$, $a_y = g$, $s_y = 11m$

$= 10 \times \frac{3}{5}$, $a_y = 10$, $s_y = 11m$

$= 6 m/s$

$\Rightarrow s_y = u_y t + \frac{1}{2} a_y t^2$

$\Rightarrow 11 = 6t + 5t^2$

$\Rightarrow 5t^2 + 6t - 11 = 0 \Rightarrow t = \frac{-6 \pm \sqrt{36 + 220}}{10}$

$t = \frac{-6 + 16}{10} = \frac{-10}{10} = 1s$

(2) $R = (u\cos\theta)T = 10\cos 37° = 10 \times \frac{4}{5} = 8m$

(3) $V_x = u\cos\theta = 10\cos 37° = 8 m/s$

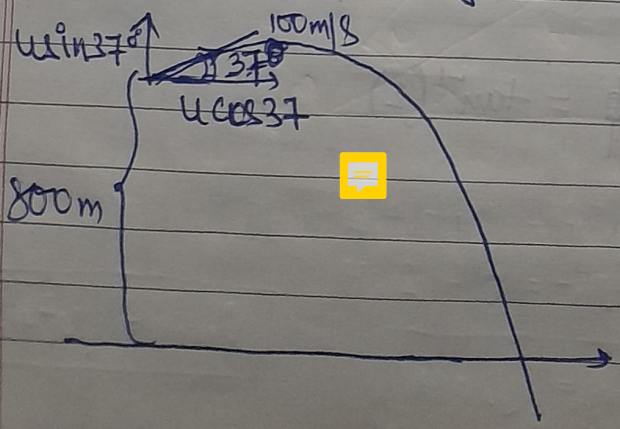
$V_y^2 = u_y^2 + 2as$

$V_y^2 = 6^2 + 2(10)(11) = 256$

$V_y = 16$

$V_{net} = \sqrt{8^2 + 16^2} = \sqrt{64 + 256} = \sqrt{320} = 8\sqrt{5}\ m/s$

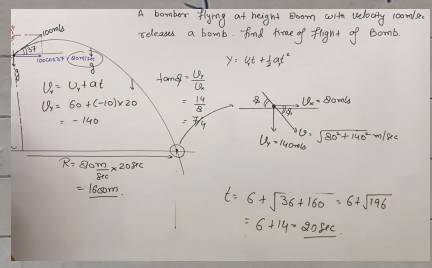
Q



A bomber flying at height 800m with $u = 100 m/s$ releases a bomb. Find Time of flight of bomb

$s_y = -800m$, $u_y = 100\sin 37°$ $a_y = -g$

$= 60 m/s$ $a_y = -10 m/s^2$



$$s_y = u_y T + \frac{1}{2} a_y T^2$$

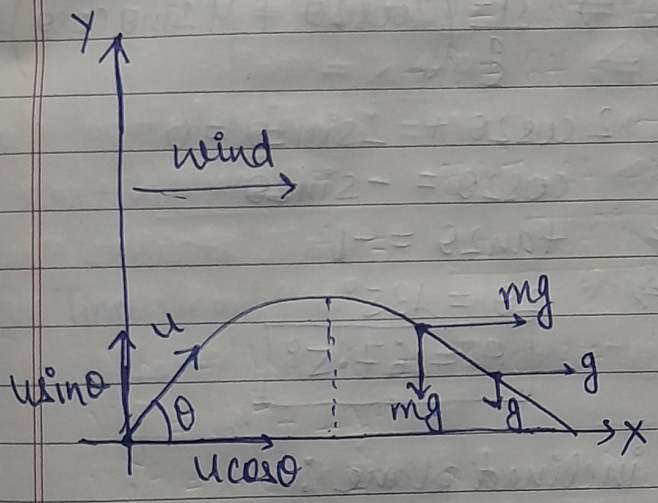
$$-800 = 60T - 5T^2$$

$$5T^2 - 60T - 800 = 0$$

$$T^2 - 12T - 160 = 0$$

$$T = \frac{12 \pm \sqrt{144 + 4(1)(160)}}{2} = \frac{12 \pm \sqrt{784}}{2} = \frac{12 + 28}{2} = 20s$$

Q Analyze the motion



X	Y
$u_x = u\cos\theta$	$v_y = u\sin\theta$
$a_x = g$	$a_y = -g$
$v_x = u_x + a_x t$	$v_y = v_y + a_y t$
$v_x = u\cos\theta + gt$	$v_y = u\sin\theta - gt$
$v_x^2 = u_x^2 + 2a_x x$	$v_y^2 = u_y^2 + 2a_y y$
$v_x^2 = u^2\cos^2\theta + 2gx$	$v_y^2 = u^2\sin^2\theta - 2gy$
$x = u_x t + \frac{1}{2}a_x t^2$	$y = u_y t + \frac{1}{2}a_y t^2$
$x = (u\cos\theta)t + \frac{1}{2}gt^2$	$y = u\sin\theta\, t - \frac{1}{2}gt^2$

① Time of flight (T):

$$y = u\sin\theta\, t - \frac{1}{2}gt^2$$

$$0 = (u\sin\theta)t - \frac{1}{2}gt^2$$

$$\boxed{T = \frac{2u\sin\theta}{g}}$$

② Maximum H: $v_y^2 = u^2\sin^2\theta - 2gy$

$$0 = u^2\sin^2\theta - 2gH$$

$$\boxed{H = \frac{u^2\sin^2\theta}{2g}}$$

③ Range (R): $R = (u\cos\theta)T + \frac{1}{2}a_x T^2$

$$R = (u\cos\theta)\left(\frac{2u\sin\theta}{g}\right) + \frac{1}{2}g\left(\frac{4u^2\sin^2\theta}{g^2}\right)$$

$$R = \frac{u^2\sin 2\theta}{g} + \frac{2u^2\sin^2\theta}{g}$$

$$\boxed{R = \frac{u^2}{g}\left(\sin 2\theta + 2\sin^2\theta\right)}$$

④ Angle θ for which range is maximum

$R = f(\theta)$

$$\frac{dR}{d\theta} = 0 \Rightarrow \frac{u^2}{g}\left(2\cos 2\theta + 4\sin\theta\cos\theta\right) = 0$$

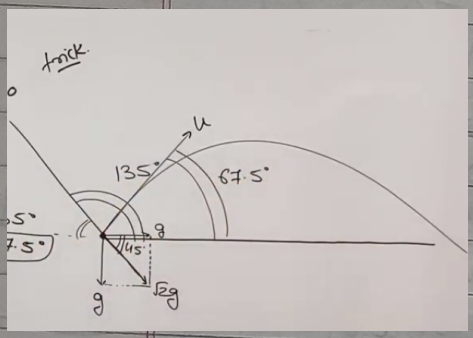
$$= 2\cos 2\theta + 2\sin 2\theta = 0$$

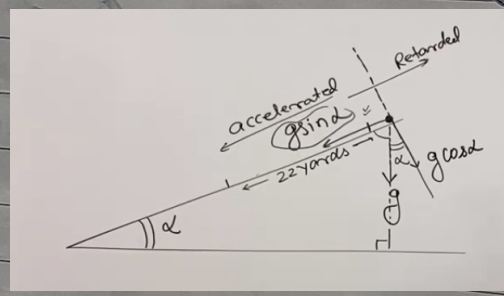
$$= \cos 2\theta = -\sin 2\theta$$

$$= \tan 2\theta = -1$$

$$\Rightarrow 2\theta = 135^\circ$$

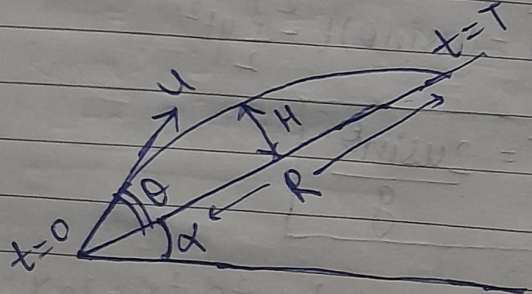
$$\boxed{\theta = 67.5^\circ}$$





Projectile Motion on inclined plane:

① Up the incline:



α = Angle of inclination

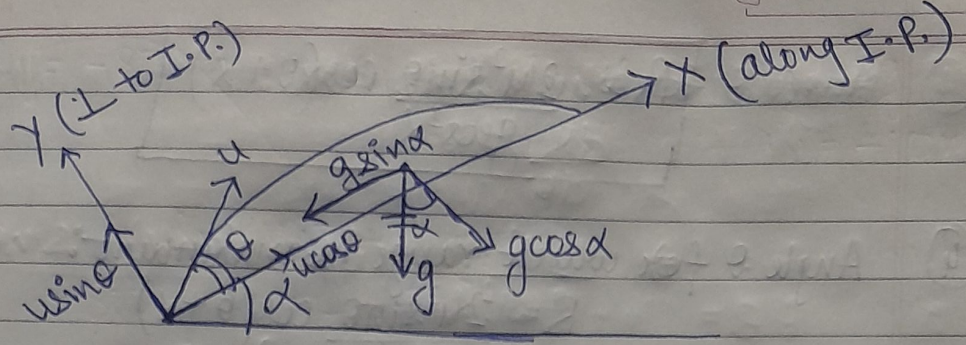
θ = Angle of projection from Inclined plane

$\theta + \alpha$ = Angle of projection from ground

T = time of flight

R = Range

h = max^m height from I.P.



X (Along I.P)	Y (⊥ to I.P)
$u_x = u\cos\theta$	$v_y = u\sin\theta$
$a_x = -g\sin\alpha$	$a_y = -g\cos\alpha$
$v_x = u_x + a_x t$	$v_y = u_y + a_y t$
$v_x = u\cos\theta - g\sin\alpha t$	$v_y = u\sin\theta - g\cos\alpha t$
$v_x^2 = u_x^2 + 2a_x x$	$v_y^2 = u_y^2 + 2a_y y$
$v_x^2 = u^2\cos^2\theta - 2g\sin\alpha x$	$v_y^2 = u^2\sin^2\theta - 2g\cos\alpha y$
$x = u_x t + \frac{1}{2}a_x t^2$	$y = u_y t + \frac{1}{2}a_y t^2$
$x = (u\cos\theta)t - \frac{1}{2}g\sin\alpha t^2$	$y = (u\sin\theta)t - \frac{1}{2}g\cos\alpha t$

① Time of flight:

$$y = (u\sin\theta)t - \frac{1}{2}g\cos\alpha t^2$$

$$0 = (u\sin\theta)t - \frac{1}{2}g\cos\alpha t^2$$

$$\boxed{t = 0, \frac{2u\sin\theta}{g\cos\alpha}}$$

$$\boxed{T = \frac{2u_y}{g_y}}$$

② Maximum height from inclined plane (H):

$$v_y^2 = u^2\sin^2\theta - 2g\cos\alpha H$$

$$0 = u^2\sin^2\theta - 2g\cos\alpha (H)$$

$$\boxed{H = \frac{u^2\sin^2\theta}{2g\cos\alpha}}$$

$$\boxed{H = \frac{u_y^2}{2g_y}}$$

③ Range (R):

$$R = u_x t + \frac{1}{2}a_x t^2$$

$$R = (u\cos\theta)\left(\frac{2u\sin\theta}{g\cos\alpha}\right) - \frac{1}{2}g\sin\alpha\left(\frac{4u^2\sin^2\theta}{g^2\cos^2\alpha}\right)$$

$$R = \frac{2u^2\sin\theta}{g\cos^2\alpha}\left(\cos\theta\cos\alpha - \sin\theta\sin\alpha\right)$$

$$\boxed{R = \frac{2u^2 \sin\theta \cos(\theta+\alpha)}{g\cos^2\alpha}}$$

(4) Angle θ for which range is maximum:

$$R = \frac{2u^2 \sin\theta \cos(\theta+\alpha)}{g\cos^2\alpha}$$

$$R = f(\theta)$$

$$\frac{dR}{d\theta} = 0 \Rightarrow \frac{2u^2}{g\cos^2\alpha} \frac{d}{d\theta}\left(\sin\theta \cos(\theta+\alpha)\right) = 0$$

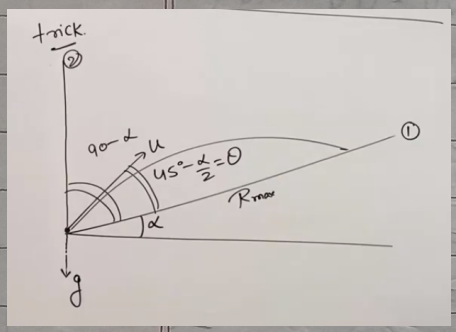
$$= -\sin\theta \sin(\theta+\alpha) + \cos(\theta+\alpha)\cos\theta = 0$$

$$= \cos(\theta+\alpha)\cos\theta - \sin\theta\sin(\theta+\alpha) = 0$$

$$= \cos(2\theta+\alpha) = 0$$

$$\Rightarrow 2\theta+\alpha = 90^\circ$$

$$\boxed{\theta = \frac{90^\circ - \alpha}{2}}$$



(5) Maximum Range:

$$R_{max} = \frac{2u^2 \sin\left(45-\frac{\alpha}{2}\right)\cos\left(45+\frac{\alpha}{2}\right)}{g\cos^2\alpha}$$

$2\sin A\cos B = \sin(A+B)+\sin(A-B)$

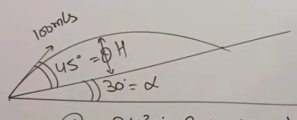
$$= \frac{u^2}{g\cos^2\alpha}\left(\sin 90 + \sin(-\alpha)\right)$$

$$= \frac{u^2}{g\cos^2\alpha}(1-\sin\alpha) = \frac{u^2}{g(1-\sin^2\alpha)}(1-\sin\alpha)$$

$$\boxed{R_{max} = \frac{u^2}{g(1+\sin\alpha)}}$$

Projectile Motion on inclined plane

Ex



Find ① T

② H_{max} from I.P

③ Range

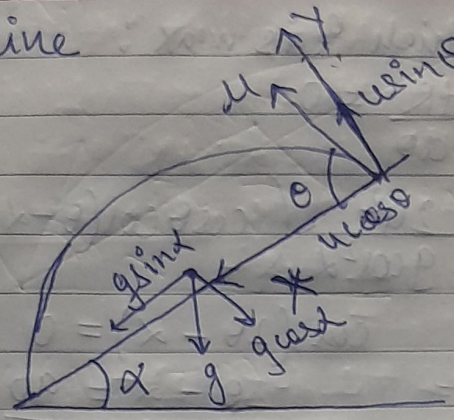
$$H = \frac{u^2\sin^2\theta}{2g\cos\alpha} = \frac{100\times100\times\frac{1}{2}}{2\times10\times\frac{\sqrt{3}}{2}} = 500\sqrt{3}\ m$$

$$T = \frac{2\times10\times\sin45^\circ}{10\times\cos30} = \frac{20}{\sqrt{3}}\times\frac{1}{\sqrt{2}}\times 2 = \frac{20\sqrt{2}}{\sqrt{3}}\ sec$$

$$R = \frac{2u^2\sin\theta\cos(\theta+\alpha)}{g\cos^2\alpha} = \frac{2\times(100)^2\sin45\,(\cos75)}{10(\cos30)}$$

$\cos75 = \frac{\sqrt{3}-1}{2\sqrt{2}}$

② Down the incline



X (Down the incline)	Y (⊥ to incline)
$u_x = u\cos\theta$	$u_y = u\sin\theta$
$a_x = g\sin\alpha$	$a_y = -g\cos\alpha$
$v_x = u_x + a_x t$	$v_y = u_y + a_y t$
$v_x = u\cos\theta + g\sin\alpha\, t$	$v_y = u\sin\theta - g\cos\alpha\, t$
$v_x^2 = u_x^2 + 2a_x x$	$v_y^2 = u_y^2 + 2a_y y$
$v_x^2 = u^2\cos^2\theta + 2g\sin\alpha\, x$	$v_y^2 = u^2\sin^2\theta - g\cos\alpha\, y$
$x = u_x t + \frac{1}{2}a_x t^2$	$y = u_y t + \frac{1}{2}a_y t^2$
$x = u\cos\theta\, t + \frac{1}{2}g\sin\alpha\, t^2$	$y = u\sin\theta\, t - \frac{1}{2}g\cos\alpha\, t^2$

① Time of flight (T):

$$y = u\sin\theta - \frac{1}{2}g\cos\alpha\, t^2$$

$$\boxed{T = \frac{2u\sin\theta}{g\cos\alpha} = \frac{2u_y}{g_y}}$$

② Maximum Height (H).

$$v_y^2 = u_y^2 + 2a_y y$$

$$v_y^2 = u^2\sin^2\theta - g\cos\alpha\, H$$

$$0 = u^2\sin^2\theta - g\cos\alpha\, H$$

$$\boxed{H = \frac{u^2\sin^2\theta}{2g\cos\alpha} = \frac{u_y^2}{2g_y}}$$

③ Range (R): $x = u\cos\theta\, t + \frac{1}{2}g\sin\alpha\, t^2$

$$x = u\cos\theta\left(\frac{2u\sin\theta}{g\cos\alpha}\right) + \frac{1}{2}g\sin\alpha\left(\frac{2u^2\sin^2\theta}{g^2\cos^2\alpha}\right)$$

$$\boxed{R = \frac{2u^2\sin\theta}{g\cos^2\alpha}\left(\cos(\theta - \alpha)\right)}$$

④ Angle θ for which R is max :

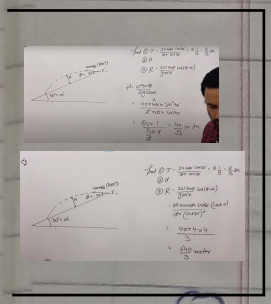
$$\frac{dR}{d\theta} = 0$$

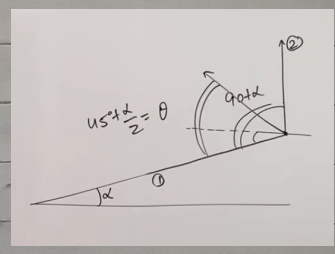
$$\Rightarrow \frac{2u^2}{g\cos^2\alpha}\left(\cos\theta\cos(\theta-\alpha) - \sin\theta\sin(\theta-\alpha)\right) = 0$$

$$= \cos(2\theta-\alpha) = 0$$

$$= 2\theta - \alpha = 90$$

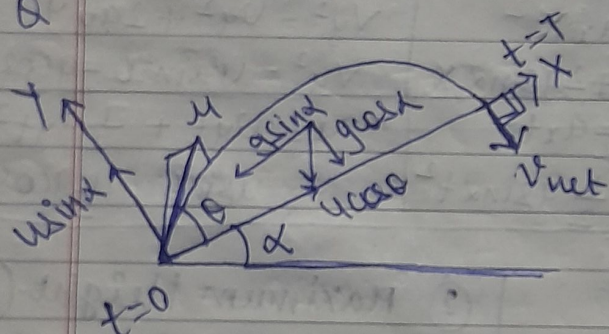
$$\boxed{\theta = \frac{90+\alpha}{2}}$$





⑤ Maximum Range : $\boxed{R_{max} = \frac{u^2}{g(1-\sin\alpha)}}$

Q



Find relation b/w θ and α such that ball strikes the inclined plane normally ($\theta = 90°$)

When ball strikes the inclined plane ⊥, $V_x = 0$

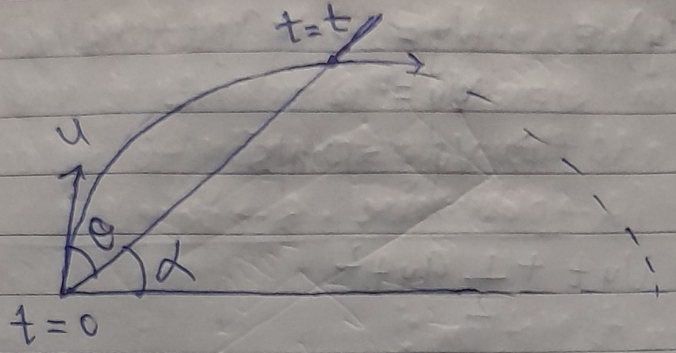
$V_y = -u\sin\alpha$

Now, $V_y = u_y + a_y t$

$$0 = u\cos\theta - g\sin\alpha \cdot \frac{2u\sin\theta}{g\cos\alpha}$$

$$\boxed{2\tan\alpha = \cot\theta}$$

Q Find Relation b/w θ and α such the projectile strikes the inclined plane horizontally.



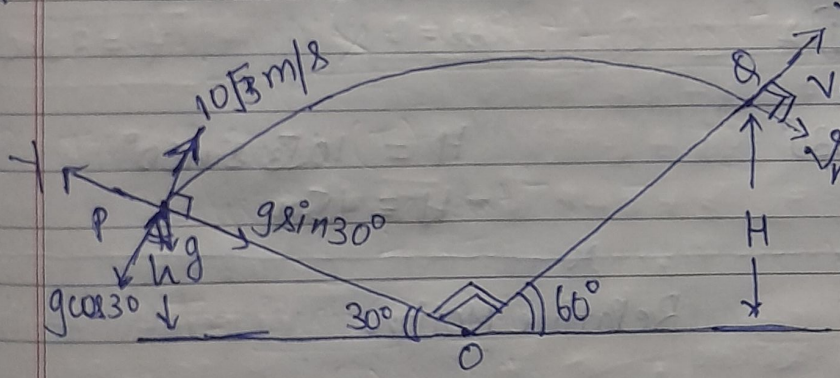
In Ground to ground projectile velocity is purely in horizontal direction at max H, or R/2.

At that time, $t = T_{I.P} = \frac{1}{2} T_{G.G}$

$$\frac{2u\sin\theta}{g\cos\alpha} = \frac{1}{2} \times \frac{2u\sin(\theta+\alpha)}{g}$$

$$= \boxed{2\sin\theta = \sin(\theta+\alpha)\cos\alpha}$$

Q

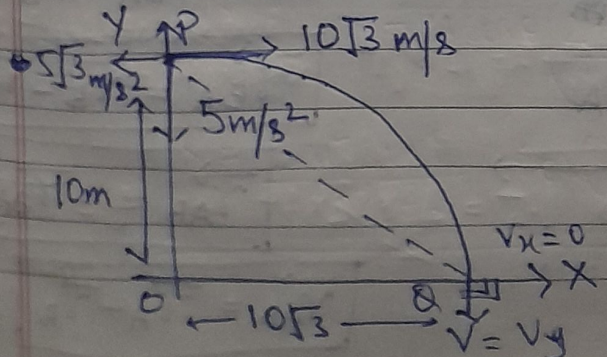


Find
① Time of flight
② Speed with which ball strikes at Q
③ Height of points P and Q
④ Distance PQ

$a_y = -g\sin30° = -g/2 = -5\ m/s$

$a_x = -g\cos30° = -g\sqrt{3}/2 = -5\sqrt{3}\ m/s$

For ease, we can rotate the diagram 60° clockwise



① Time of flight (T):

$V_x = u_x + a_x t$

$0 = 10\sqrt{3} - 5\sqrt{3}t$

$\boxed{T = 2s}$

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(2) Speed (v_y) = $v_y = u_y + a_y t$

$= v_y = 0 - 5 \times 2$

$\Rightarrow v_y = -10 m/s$

(b) $PQ = u_y t + \frac{1}{2} a_y t^2$

$PQ = 0 \times 2 + \frac{1}{2}(-5)(2)^2$

$PQ = -10m$

distance $h = 10m$

$OQ = u_x t + \frac{1}{2} a_x t^2$

$OQ = 10\sqrt{3} \times 2 - \frac{1}{2} 5\sqrt{3} \times 2^2 = 10\sqrt{3}$

$OQ = 10\sqrt{3} m$

(c) $PQ = \sqrt{10^2 + (10\sqrt{3})^2} = 20m$

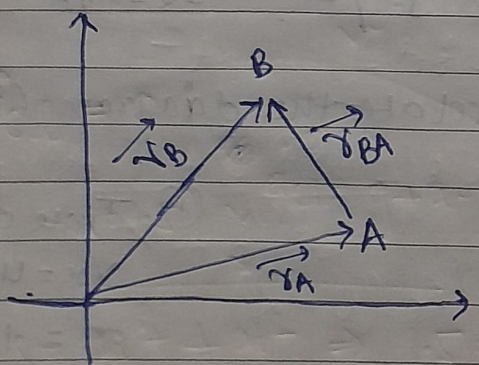
(3) Refer Fig (1)

$h = 10 \sin 30°$ $\qquad$ $H = 10\sqrt{3} \sin 60°$

$h = 5m$ $\qquad$ $H = 15m$

Relative Motion

Relative Position:



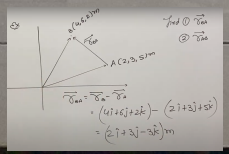
$\vec{r}_{BA}$ = Position of B wrt A

$\vec{r}_A + \vec{r}_{BA} = \vec{r}_B$

$$\boxed{\vec{r}_{BA} = \vec{r}_B - \vec{r}_A}$$

$$\boxed{\vec{r}_{AB} = -\vec{r}_{BA}}$$

$$\boxed{\vec{r}_{AB} = \vec{r}_A - \vec{r}_B}$$



Relative Velocity : Rate of change of relative position.

$$\boxed{\vec{V}_{rel} = \frac{d}{dt}(\vec{r}_{rel})}$$

$\vec{V}_{AB}$ = Velocity of A wrt B = $\frac{d}{dt}(\vec{r}_{AB}) = \frac{d}{dt}(\vec{r}_A - \vec{r}_B)$

$\vec{V}_{AB}$ = Velocity of A wrt B. = $\vec{V}_A - \vec{V}_B = \vec{V}_A + (-\vec{V}_B)$

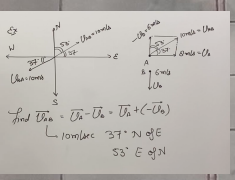
$\vec{V}_{BA}$ = Velocity of B wrt A = $\vec{V}_B - \vec{V}_A$
$= \vec{V}_B + (-\vec{V}_A)$

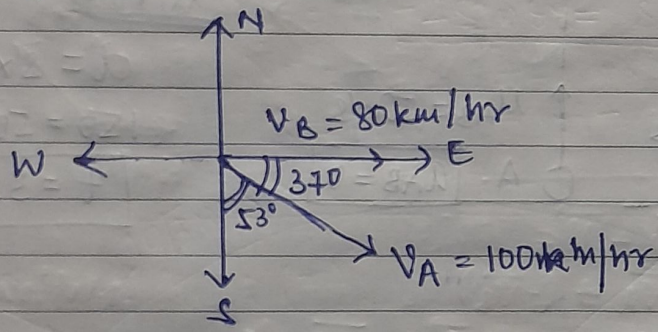
$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$ $\quad$ $\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$

$\vec{V}_{AB} = -\vec{V}_{BA}$

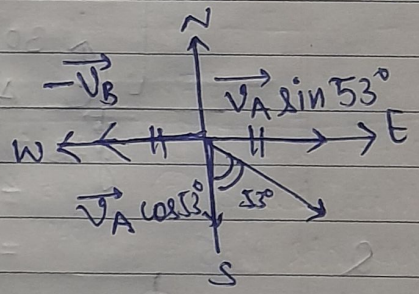
$\vec{V}_{AA} = \vec{V}_A - \vec{V}_A = \vec{0}$

Q Ship A is moving with velocity 100 km/hr 37° S of E. Ship B is moving with velocity 80 km/hr toward east. Find velocity of ship A wrt Ship B.





$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$
$= \vec{V}_A + (-\vec{V}_B)$



$\vec{V}_A \sin 53° = 100 \times \frac{4}{5} = 80$ km/hr

$\vec{V}_A \cos 53° = 100 \times \frac{3}{5} = 60$ km/hr

$\therefore \vec{V}_{AB} = 60$ km/hr towards south

Relative acceleration : Rate of change of relative velocity.

$$\vec{a}_{rel} = \frac{d}{dt}(\vec{v}_{rel})$$

$\vec{a}_{AB} = \vec{a}_A - \vec{a}_B$ $\quad$ $\vec{a}_{BA} = \vec{a}_B - \vec{a}_A$

$\vec{a}_{AB} = \vec{a}_A - \vec{a}_B$ (dt, dt)
$= \vec{a}_A + (-\vec{a}_B)$

$\vec{a}_{BA} = \vec{a}_B - \vec{a}_A$
$= \vec{a}_B + (-\vec{a}_A)$

$\vec{a}_{AB} = -\vec{a}_{BA}$ → opposite

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VVIP Note: If we want to study relative motion of B wrt A then reverse the velocity of B and acceleration of A and add them vectorially to velocity and acc^n of B. (A → observer, B → object)

Equation of Motion:

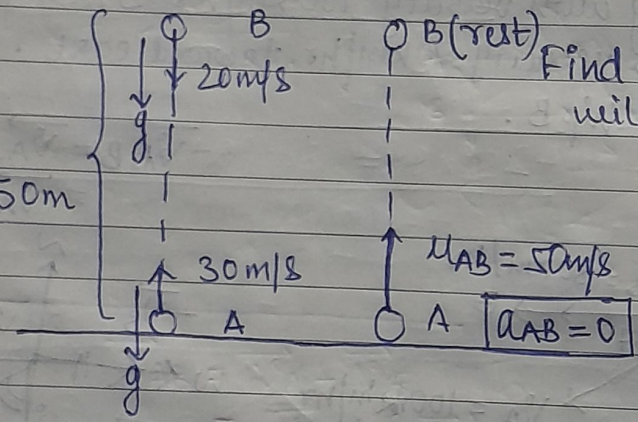
$$\vec{a}_{rel} = \text{const}$$

(1) $\vec{v}_{rel} = \vec{u}_{rel} + \vec{a}_{rel} t$

(2) $\vec{s}_{rel} = \vec{u}_{rel} + \frac{1}{2}\vec{a}_{rel} t^2$

(3) $\vec{v}^2_{rel} = \vec{u}^2_{rel} + 2\vec{a}_{rel}\vec{s}_{rel}$

Q



Find time after which the balls will collide.

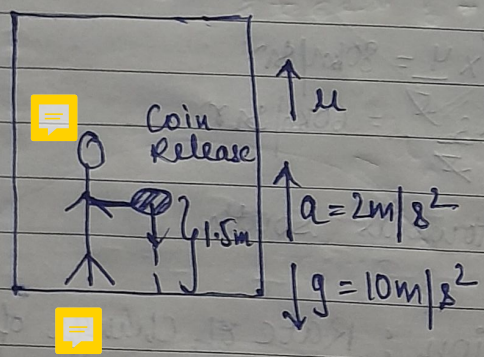
wrt B

$d = s \times t$

$150 = 50 \times t$

$\boxed{t = 3 \text{ sec}}$

Q

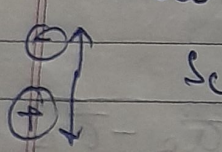


Find time taken by coin to strike the floor.

wrt lift

If we observe wrt lift, lift is going up with velocity u and boy is inside it but for boy velocity of lift will be zero

$\therefore \vec{u}_{cl} = 0$



$s = ut + \frac{1}{2}at^2 \Rightarrow 1.5 = \frac{1}{2} \times 12 \times t^2 \Rightarrow t = \sqrt{\frac{1.5}{6}}$ s

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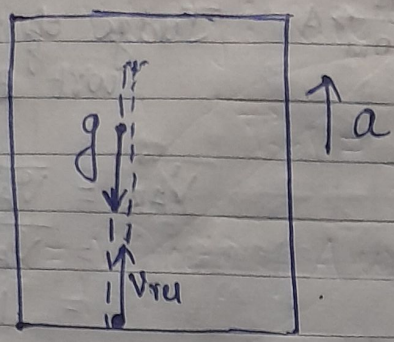
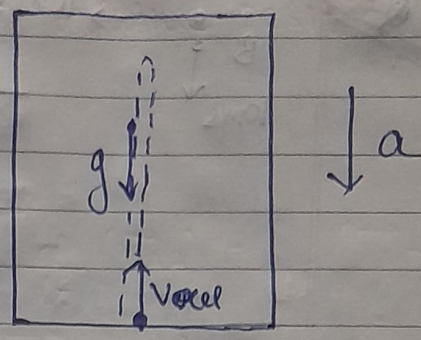
Note

① 

$a_{rel} = g + a$

$T = \frac{2u_{rel}}{g+a}$

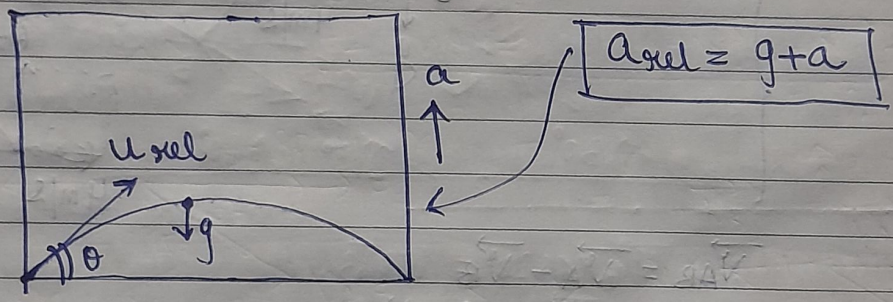
$H = \frac{u^2_{rel}}{2(g+a)}$

② 

$a_{rel} = g - a$

$T = \frac{2u_{rel}}{g-a}$

$H = \frac{u^2_{rel}}{2(g-a)}$

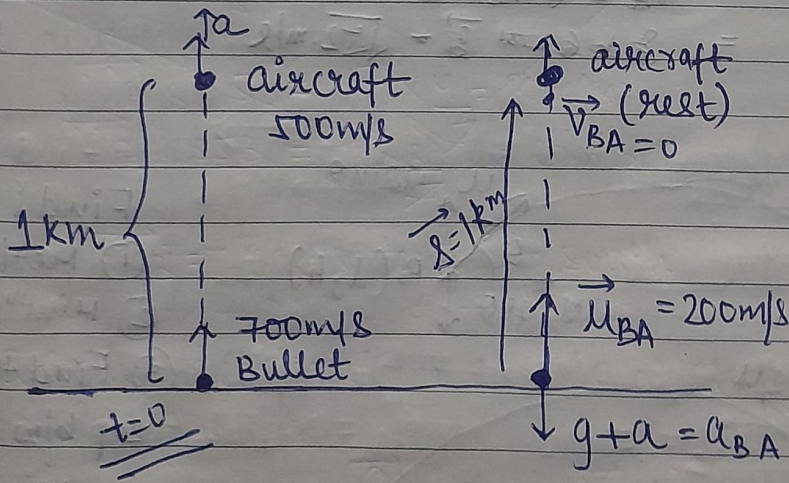
③ 

$a_{rel} = g + a$

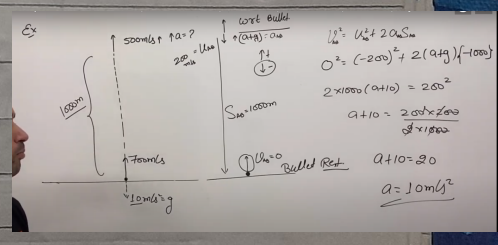
$T = \frac{2u_{rel}\sin\theta}{g+a}$

$H = \frac{u^2_{rel}\sin^2\theta}{2(g+a)}$

$R = \frac{u^2_{rel}\sin 2\theta}{(g+a)}$

Q 

Find the min^m acc^n of aircraft in vertical direction such that it just escapes the bullet.



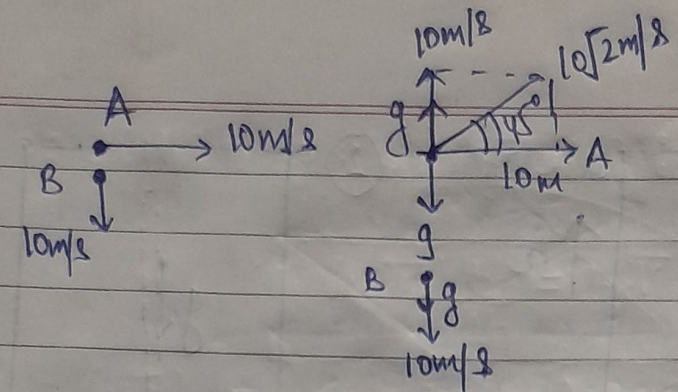
wrt aircraft

$V^2_{BA} = u^2_{BA} = 2a_{BA}s_{BA}$

$0 = (200)^2 - 2(10+a)\times1000$

$10 + a = \frac{200\times200}{2\times1000} \Rightarrow$ $\boxed{a = 10 m/s^2}$

Q

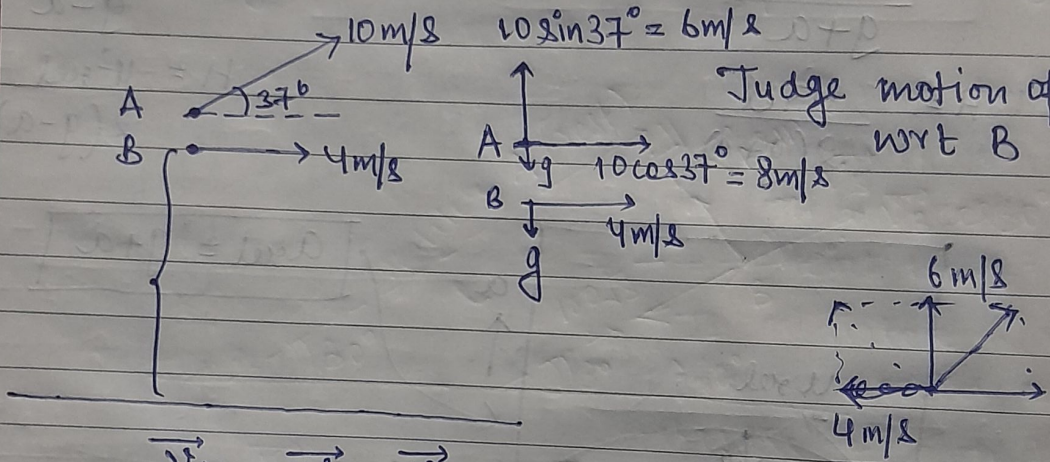


Judge of motion of A wrt B.

$$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$$
$$= \vec{V}_A + (-\vec{V}_B)$$
$$= 10\sqrt{2}\ m/s$$

Motion of A wrt B will be straight line and acc^n = 0 ; V = const.

Q

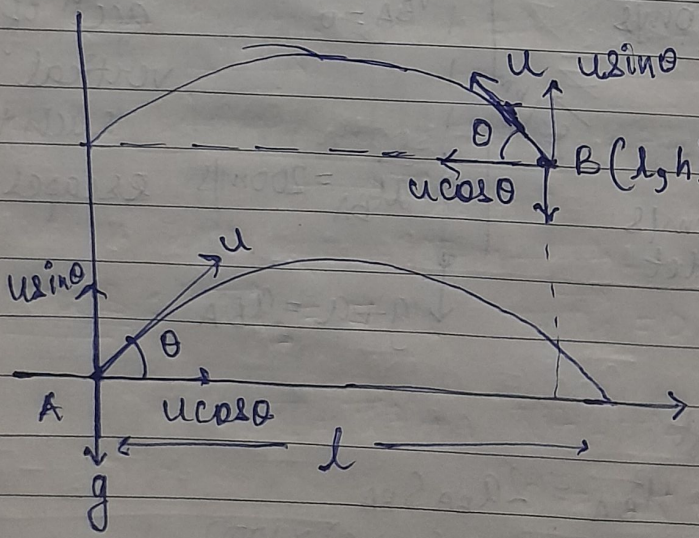


Judge motion of A wrt B

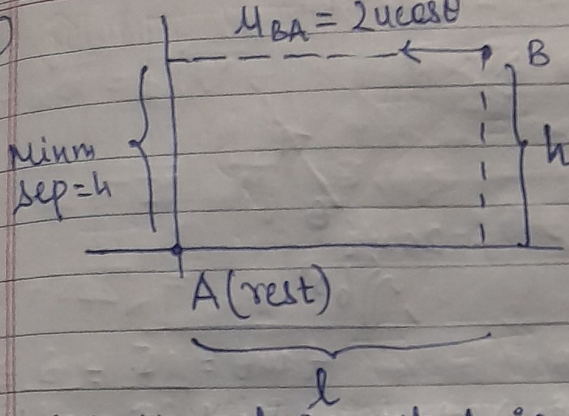
$$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$$
$$= \vec{V}_A + (-\vec{V}_B)$$
$$= \sqrt{52}\ m/s$$

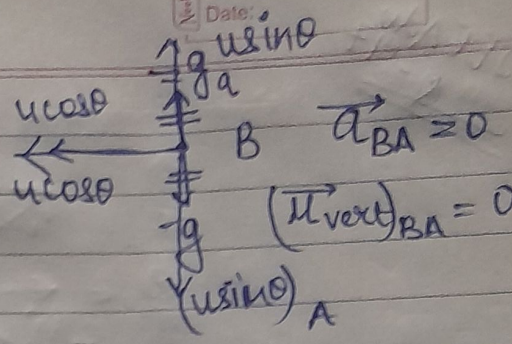
Motion is straight line

Q



Find:
1. Motion of B wrt A
2. Min^m sep b/w A & B
3. Find the time when sep b/w A and B is minm

① 



$(\vec{u}_{Hori})_{BA} = 2u\cos\theta$

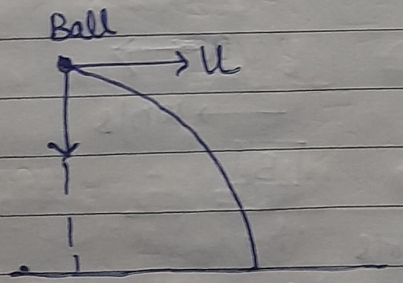
∴ Motion of B wrt A is straight line

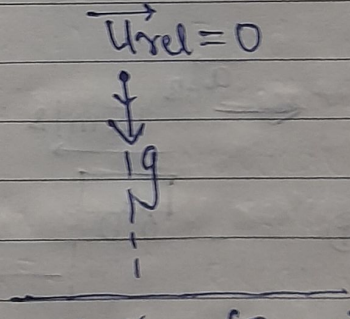
② Min^m sep = h

③ $t = \dfrac{l}{2u\cos\theta}$

Note: Any two objects moving under gravity will have $\vec{a}_{rel} = 0$ and hence $\vec{v}_{rel}$ = const and then their relative motion will be uniform st. line motion.

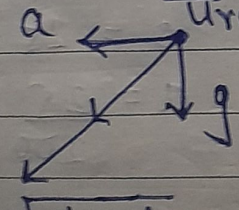
Ex Suppose A is in a train moving with const velocity u and B is standing on platform. A drops a ball from the window.





Parabolic path (For B) St. line (For A)

→ Now, lets suppose train was accelerating with acc^n 'a'. For B the path will still remain parabola because only one force mg will be acting on ball ⇒ accn 'g'.

For A: 

Motion is st. line.

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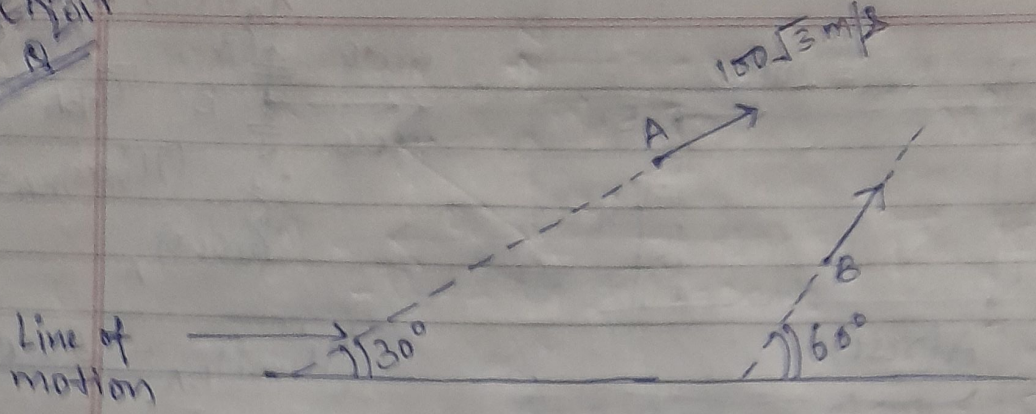
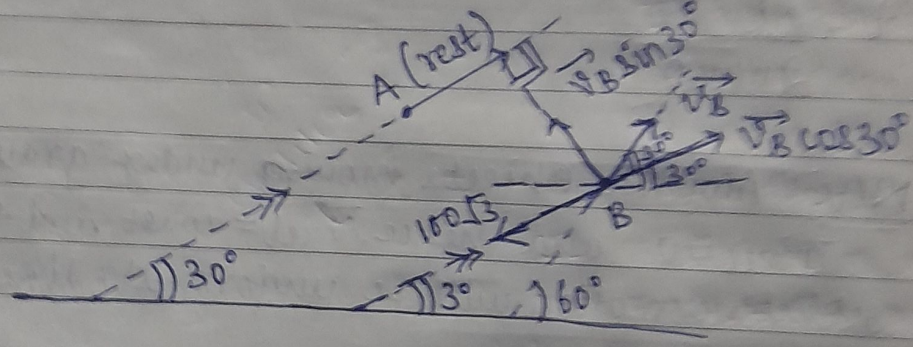


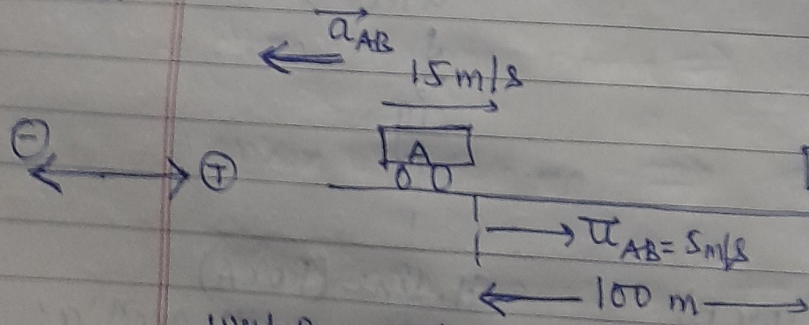
An observer in aircraft A observes that aircraft B is moving ⊥ to line of motion of A. Find $\vec{V}_B$.



$\therefore \vec{V}_B \cos 30° = 100\sqrt{3}$

$\vec{V}_B = 200$ m/s

Q



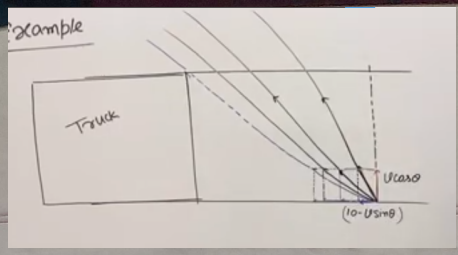
Find minm value of a so that car just avoid collision

wrt B: $\vec{u}_{AB} = 5$ m/s $\quad \vec{v}_{AB} = 0 \quad \vec{s}_{AB} = 100$

$\vec{s} = \vec{u}t + \frac{1}{2}at^2 \quad \vec{v}^2 = \vec{u}^2 + 2\vec{a}\vec{s}$

$0 = 25 + 2 \times a \times 100$

$a = \frac{25}{200} = 12.5$ cm/sec^2



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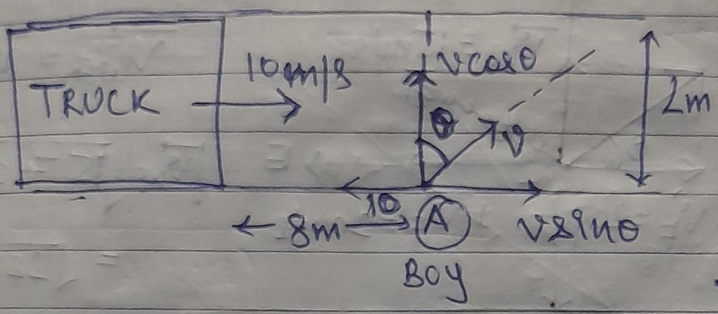
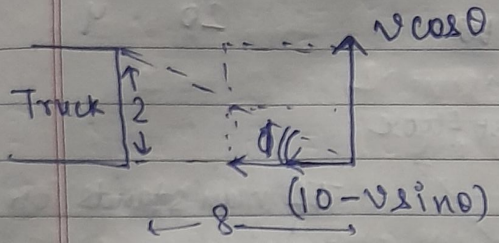


Find angle θ for which value of v is least. Also find min value of v for which boy just crosses the road w/o striking the truck.

wrt Truck:



$$\tan\phi = \frac{1}{4} = \frac{v\cos\theta}{10 - v\sin\theta}$$

$$\Rightarrow 10 - v\sin\theta = 4v\cos\theta$$

$$\Rightarrow v(4\cos\theta + \sin\theta) = 10$$

$$v = \frac{10}{4\cos\theta + \sin\theta}$$

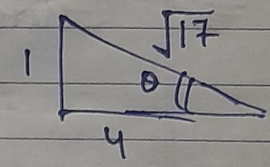
For min^m v, $4\cos\theta + \sin\theta$ should be max,

$$f(\theta) = 4\cos\theta + \sin\theta$$

$$f'(\theta) = -4\sin\theta + \cos\theta = 0$$

$$4\sin\theta = \cos\theta$$

$$\tan\theta = \frac{1}{4}$$



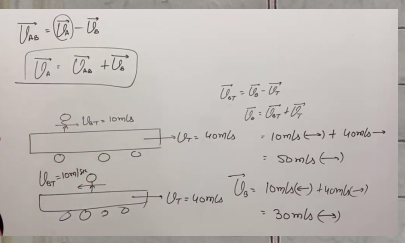
$$\boxed{\theta = \tan^{-1}\left(\frac{1}{4}\right)}$$

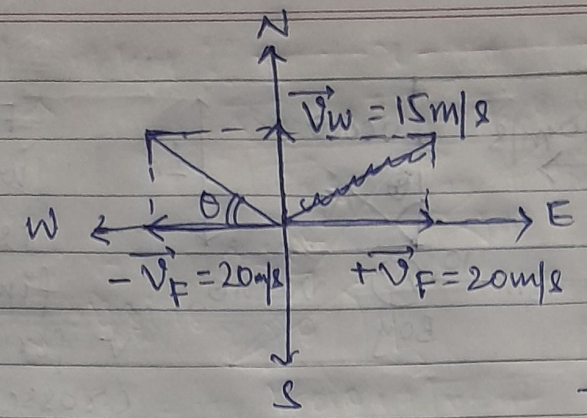
$$f(\theta)\Big|_{max} = 4\times\frac{4}{\sqrt{17}} + \frac{1}{\sqrt{17}} = \sqrt{17} \Rightarrow \boxed{v_{min} = \frac{10}{\sqrt{17}}\ m/s}$$

Fluttering of Flag: Flag flutters in direction of velocity of wind wrt flag.

$$\vec{V}_{WF} = \vec{V}_W + \vec{V}_F$$

Q Wind is blowing with speed 54km/hr along North and a car is travelling with speed 72km/hr toward east. A flag is mounted on car. Find the direction in which flag will flutter.





$\vec{v}_{wF} = \vec{v}_w - \vec{v}_F$

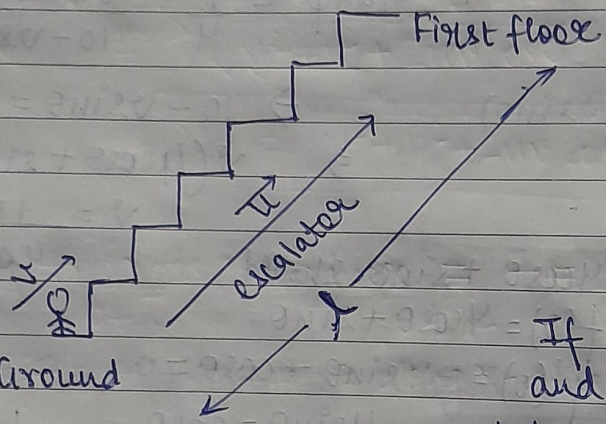
$|\vec{v}_{wF}| = \sqrt{15^2 + 20^2}$

$= \sqrt{225 + 400} = 25$

$\tan\theta = \frac{15}{20} = \frac{3}{4} \Rightarrow \theta = 37°$

∴ Flag will flutter in 37° N of W

Q



If man stands on moving escalator, it takes 10 s to reach the 1st floor.

If escalator is not moving and man walks up, it takes 15 s to reach 1st floor.

If man walks up on moving escalator, how much time he will take to reach 1st floor.

When escalator moves $\Rightarrow \frac{l}{u} = 10$

When man moves $\Rightarrow \frac{l}{v} = 15$

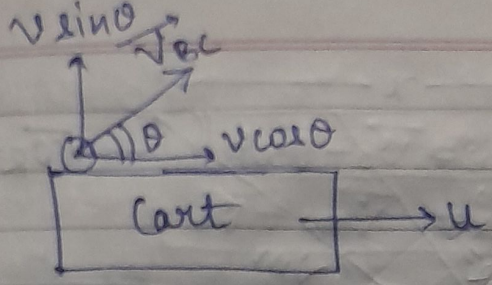
When both moves $\Rightarrow \vec{v}_B = \vec{v}_{BE} + \vec{v}_E$

$\vec{v}_B = v + u$

$\Rightarrow \frac{l}{v+u} = t \Rightarrow \frac{l}{l/15 + l/10} = t$

$t = \frac{10 \times 15}{25} = 6 s$

Note

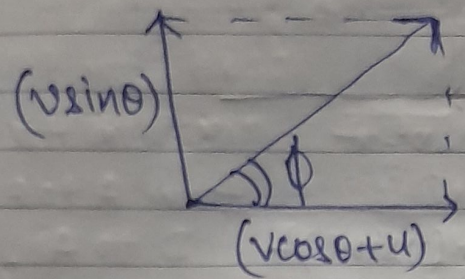


$\vec{v}_{BC} = \vec{v}$

$\vec{v}_B = \vec{v}_{BC} + \vec{v}_C$

$v_B = \sqrt{(v\sin\theta)^2 + (v\cos\theta + u)^2}$

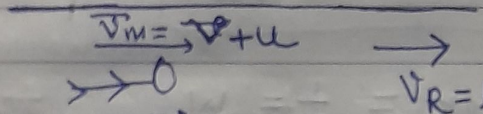
$= \sqrt{u^2 + v^2 + 2uv\cos\theta}$



$\tan\phi = \frac{v\sin\theta}{v\cos\theta + u}$

River and Swimmer Problems:

In 1-D

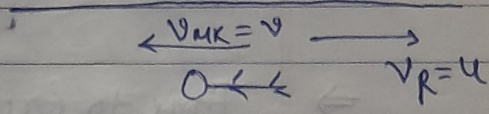


$\vec{v}$ = velocity of man in still water

downstream

$\vec{v}_{MR} = \vec{v}_M - \vec{v}_R$

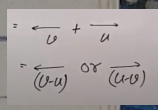
$\boxed{\vec{v}_M = \vec{v}_{MR} + \vec{v}_R}$



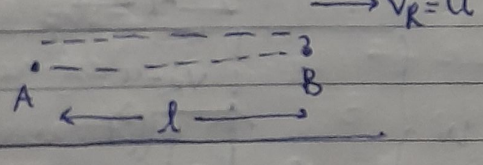
upstream

$\vec{v}_{MR} = \vec{v}_M - \vec{v}_R$

$\boxed{\vec{v}_M = \vec{v}_{MR} + \vec{v}_R}$



Q



Find time taken by the swimmer for this round trip.

$v_{MR} = v$

$t_{down} = \frac{l}{v+u}$ $\quad t_{up} = \frac{l}{v-u}$

$t = t_d + t_u = \frac{l}{v+u} + \frac{l}{v-u}$

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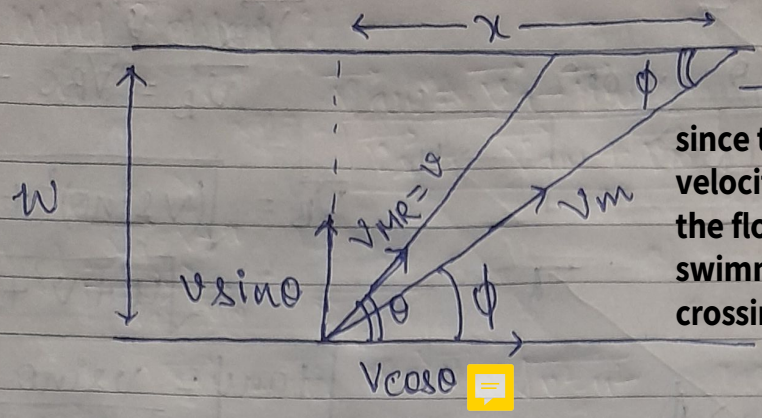
In 2-D



since there is no component of velocity of the river perpendicular to the flow of river, hence, it is only the swimmer whose velocity helps in crossing the river

$$|\vec{v_m}| = \sqrt{(v\cos\theta + u)^2 + (v\sin\theta)^2}$$
$$= \sqrt{v^2 + u^2 + 2uv\cos\theta}$$

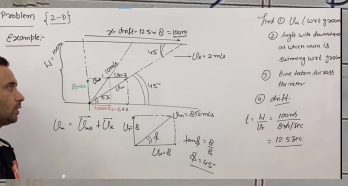
$$\boxed{v_m = \sqrt{v^2 + u^2 + 2uv\cos\theta}}$$

$$\boxed{\tan\phi = \frac{v\sin\theta}{v\cos\theta + u}}$$

$\Rightarrow$ Time to cross the river $= t = \frac{w}{v\sin\theta}$

$\Rightarrow$ Drift = displacement along X-axis $= x = v_x \times t$

$$x = \frac{(v\cos\theta + u)w}{v\sin\theta}$$

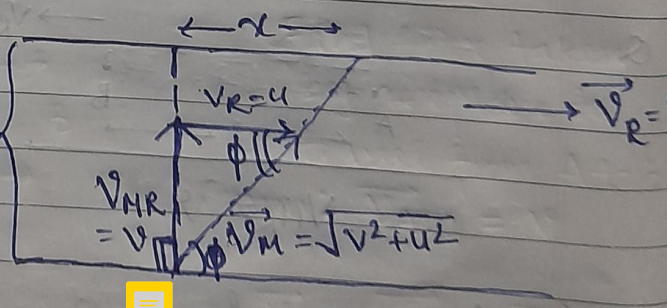


Minimum time to cross the river:

$$\downarrow \boxed{t = \frac{w}{v\sin\theta}} \uparrow$$

At $\theta = 90°$, $\sin\theta = 1$

$$\boxed{t_{min} = \frac{w}{v}}$$



$$\vec{v}_M = \vec{v}_{MR} + \vec{v}_R$$
$$= \sqrt{v^2 + u^2}$$

$\therefore \theta = 90° \Rightarrow$ Min time to cross river

$$\boxed{\tan\phi = \frac{v}{u} = \frac{v_{MR}}{v_R}}$$

$$\boxed{x = \text{drift} = v_x \times t_{min} = \frac{u \times w}{v}}$$

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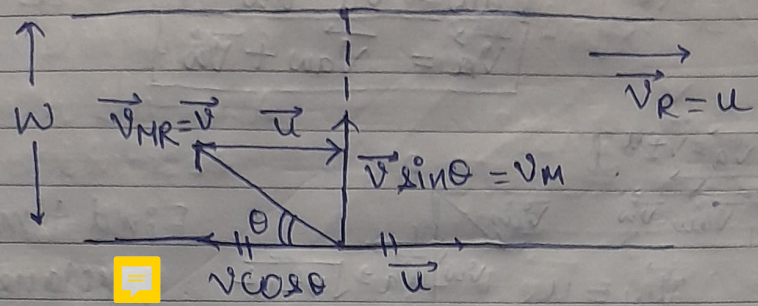
Zero drift:



For zero drift: $v\cos\theta = u$

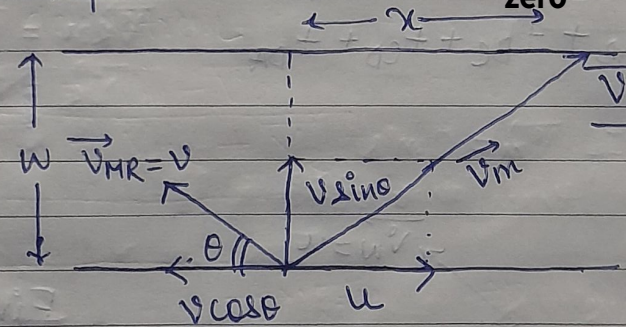
$$\cos\theta = \frac{u}{v} = \frac{v_R}{v_{MR}}$$

$$v_M = v\sin\theta = v\sqrt{1-\cos^2\theta} = \sqrt{v^2-u^2}$$

$$t = \frac{w}{\sqrt{v^2-u^2}}$$

Note: Zero drift is only possible if $v_{MR} > v_R$

Minimum Drift: when $v_R > v_{MR}$ drift cannot be zero



$v_R > v_{MR}$

$v_y = v\sin\theta$

$v_x = u - v\cos\theta$ $\quad (u > v)$

$$t = \frac{w}{v\sin\theta}$$

$$x = v_x \times t = (u - v\cos\theta)\frac{w}{v\sin\theta}$$

$$x = \frac{uw}{v}\operatorname{cosec}\theta - w\cot\theta$$

For minm drift: $\frac{dx}{d\theta} = 0 \Rightarrow \frac{-uw}{v}\operatorname{cosec}\theta\cot\theta + w\operatorname{cosec}^2\theta = 0$

$$\Rightarrow \frac{uw}{v}\frac{\cos\theta}{\sin\theta} = \frac{1}{\sin\theta} \Rightarrow \cos\theta = \frac{v}{u} = \frac{v_{MR}}{v_R}$$

At this θ, we get min. drift

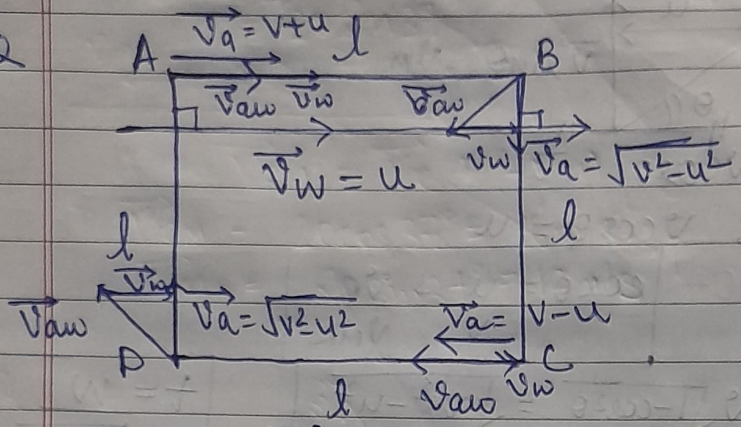
instead of river, we have wind and instead of boat/swimmer we have aeroplane

Wind Problem:

$$\vec{V}_{aw} = \vec{V}_a - \vec{V}_w$$

$$\boxed{\vec{V}_a = \vec{V}_{aw} + \vec{V}_w}$$

Q



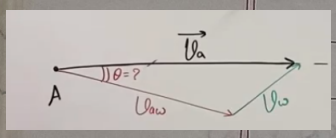
Find time taken by aeroplane to travel from A to B to C to D to A.

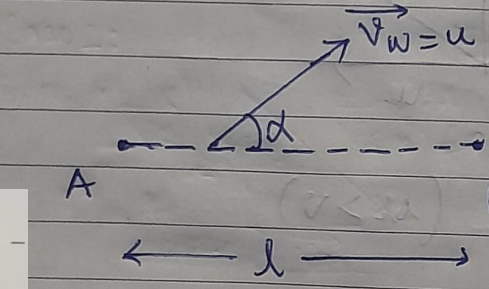
Given = $\vec{V}_{AW} = v$

$$t_{AB} = \frac{l}{v+u}, \quad t_{BC} = \frac{l}{\sqrt{v^2-u^2}}, \quad t_{CD} = \frac{l}{v-u}, \quad t_{DA} = \frac{l}{\sqrt{v^2-u^2}}$$

$$t = t_{AB} + t_{BC} + t_{CD} + t_{DA}$$

Q

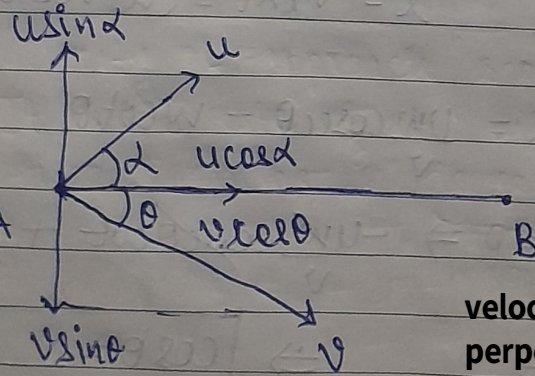




Find time taken by aeroplane to travel from A to B

Given: $\vec{V}_{AW} = v$

$$\vec{V}_A = \vec{V}_{AW} + \vec{V}_W$$



To go from A to B

$$u\sin\alpha = v\sin\theta$$

$$\sin\theta = \frac{u\sin\alpha}{v}$$

$$\cos\theta = \sqrt{1 - \frac{u^2\sin^2\alpha}{v^2}}$$

velocity perpendicular to AB must be zero for the plane to reach B

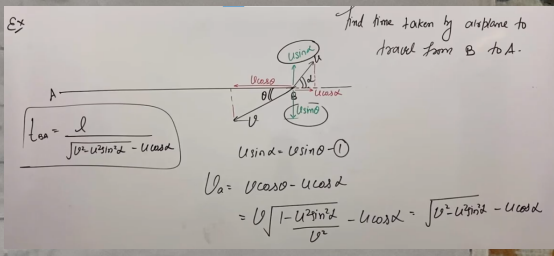
$$\vec{V_A} = V\cos\theta + u\cos\alpha$$

$$= V\sqrt{1 - \frac{u^2\sin^2\alpha}{V^2}} + u\cos\alpha$$

$$\vec{V_A} = \sqrt{V^2 - u^2\sin^2\alpha} + u\cos\alpha$$

$$\boxed{t = \frac{l}{\sqrt{V^2 - u^2\sin^2\alpha} + u\cos\alpha}}$$

Ex Find time taken by airplane to travel from B to A.



$t_{BA} = \frac{l}{\sqrt{V^2 - u^2\sin^2\alpha} - u\cos\alpha}$

$u\sin\alpha = V\sin\theta$ — (1)

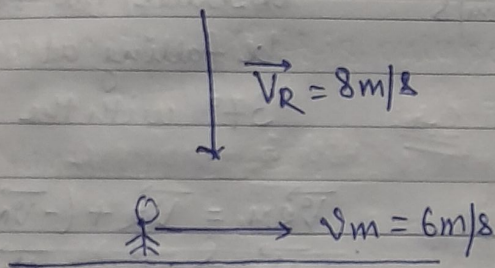
$V_a = V\cos\theta - u\cos\alpha$

$= V\sqrt{1 - \frac{u^2\sin^2\alpha}{V^2}} - u\cos\alpha = \sqrt{V^2 - u^2\sin^2\alpha} - u\cos\alpha$

Rain Problems:

$$\boxed{\vec{V_{RM}} = \vec{V_R} - \vec{V_M}}$$

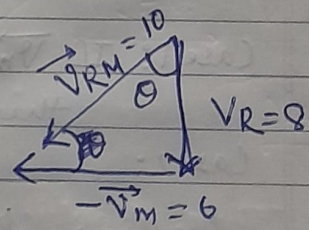
Q



Find V_{RM}
Also find direcⁿ in which man should hold the umbrella

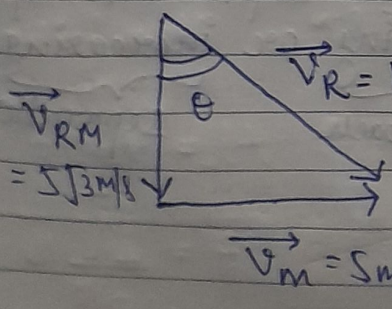
$$\vec{V_{RM}} = \vec{V_R} + (-\vec{V_M})$$

$$\vec{V_{RM}} = 10\,m/s$$



$$\tan\theta = \frac{6}{8} \Rightarrow \theta = 37°$$

Q A man running horizontally with velocity 5m/s, observes that rain is falling vertically with velocity $5\sqrt{3}$ m/s. Find actual velocity of rain.



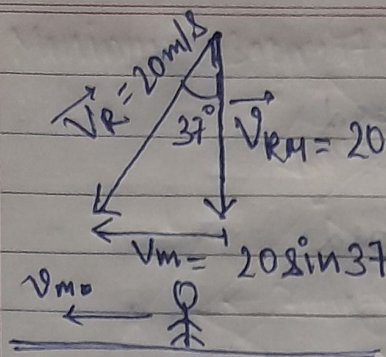
$$\vec{V_R} = \vec{V_{RM}} + \vec{V_M}$$

$$\vec{V_R} = \sqrt{5^2 + (5\sqrt{3})^2}$$

$$\vec{V_R} = 10\,m/s$$

$$\tan\theta = \frac{5}{5\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30°$$

Q



Find V_m such that he observes the rain to be falling vertically downwards.

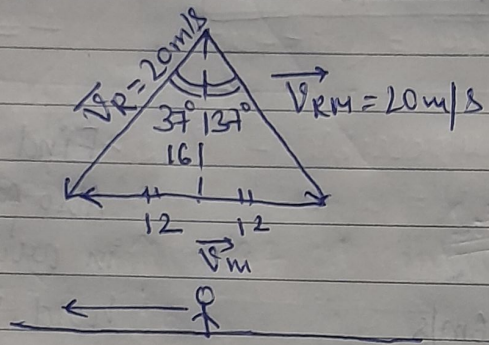
Also find velocity of rain observed by man.

$$\vec{V}_{RM} = \vec{V}_R + (-V_M)$$

$$\vec{V}_{RM} = 20\cos 37° = 16 m/s$$

$$\vec{V}_M = 20\sin 37° = 12 m/s$$

Q



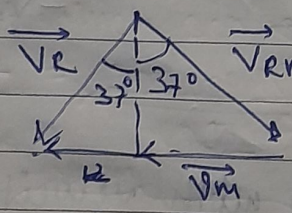
Find V_m such that he observes the rain to be falling at an angle 37° with the vertical

$$\vec{V}_{RM} = \vec{V}_R + (-\vec{V}_M)$$

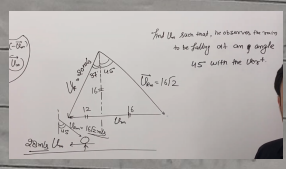
$$\vec{V}_R = \vec{V}_{RM} + \vec{V}_M$$

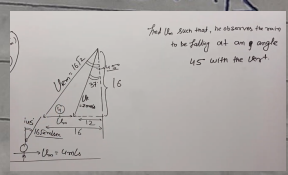
Case 1: If $\vec{V}_m = 0 m/s$
then angle = 37°

Case 2:



$$\vec{V}_m = 24 m/s$$





imp. question

Velocity of approach / separation:

It is the rate at which separation b/w two particles is decreasing or increasing.

decreasing → approach

increasing → separation

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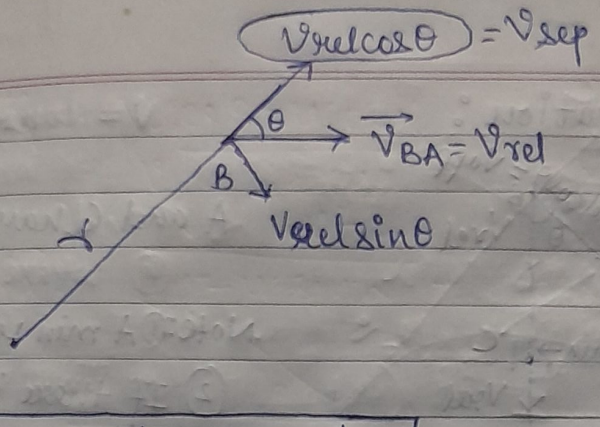
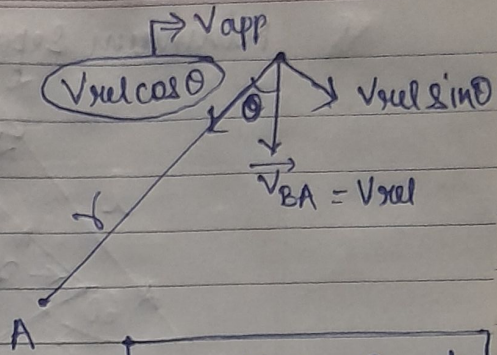


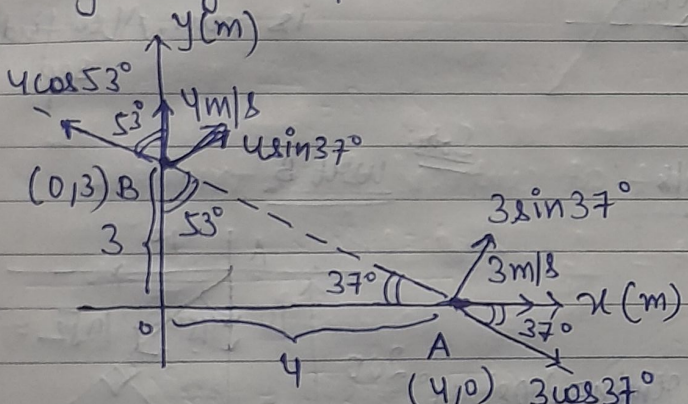


$$V_{sep} = V_{rel}\cos\theta = \frac{dr}{dt}$$

$$V_{app} = V_{rel}\cos\theta = \frac{-dr}{dt}$$

∴ V_{sep} or V_{app} is component of V_{rel} along the line joining the particles.

Q

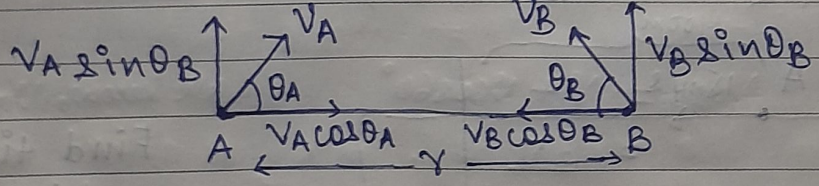


Find V_{sep} at $t=0$

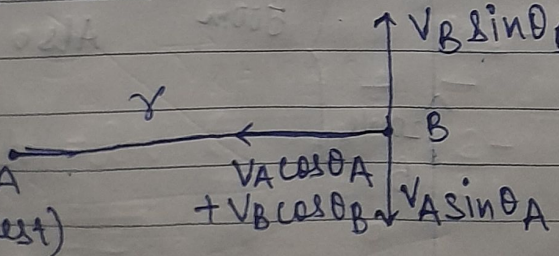
at $t=1$ sec find V_{sep}.

$$V_{sep} = 4\cos53° + 3\cos37°$$
$$= 4\times\frac{3}{5} + 3\times\frac{4}{5}$$
$$= \frac{24}{5}\ m/s$$

Condition to collide :



Now, wrt A



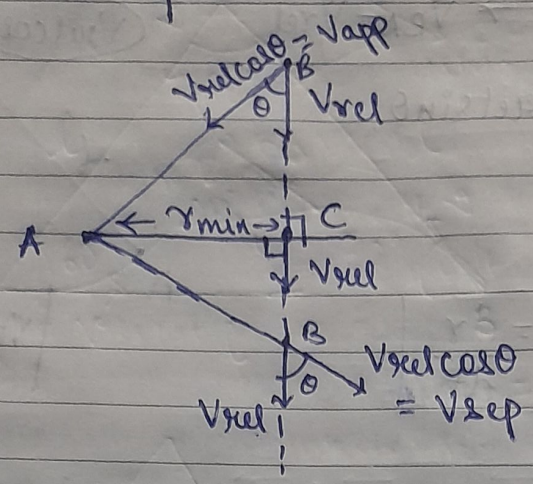
For colliding :

$$V_B\sin\theta_B = V_A\sin\theta_A$$

Time at which they collide :

$$t = \frac{r}{V_A\cos\theta_A + V_B\cos\theta_B}$$

Minimum Separation:



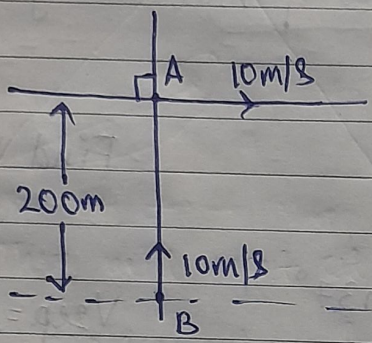
A and C have min sep.

Note: ① At min sep, Vapp | Vsep = 0

② If there is $\vec{V}_{rel}$, it should be ⊥ to $\vec{r}$

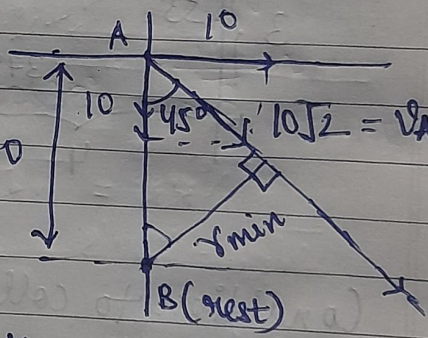
$$t = \frac{\vec{r}_{AB} \cdot \hat{v}_{AB}}{|\vec{v}_{AB}|}$$

Q



Find time at which sep is min. Also find min sep.

Soln: wrt B



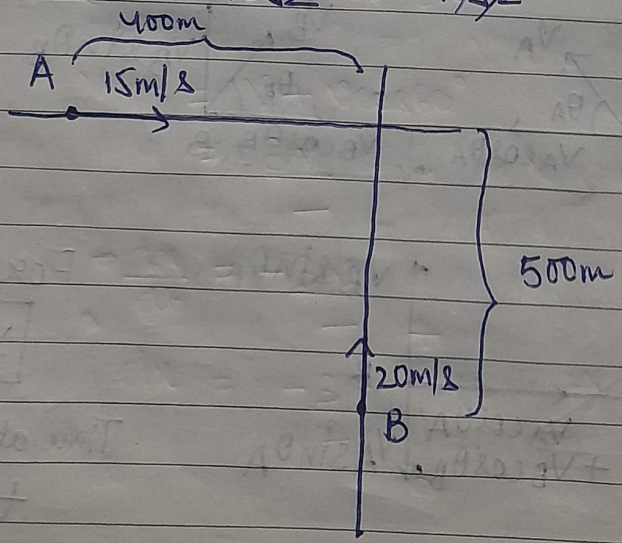
$\therefore r_{min} = 200 \sin 45°$

$= 100\sqrt{2}\ m$

Time at which sep is min

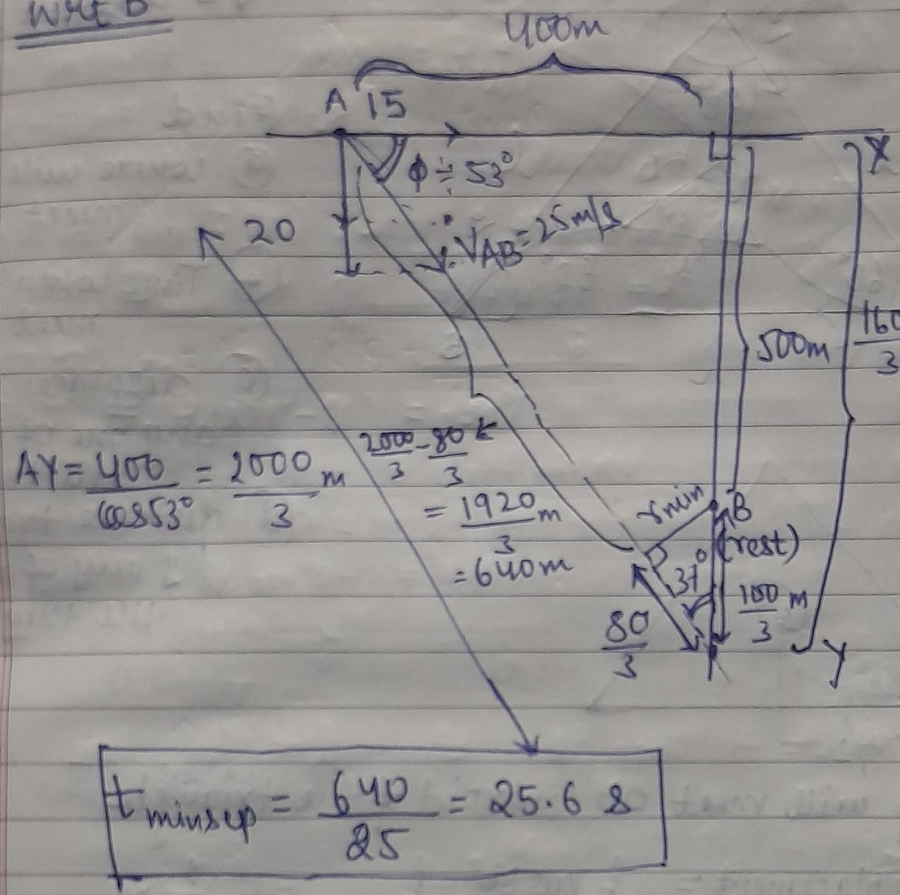
$$t = \frac{200 \cos 45°}{10\sqrt{2}} = \frac{100\sqrt{2}}{10\sqrt{2}} = 10s$$

Q



Find time at which sep is min. Also find minm sep

Wrt B



$\tan\phi = \frac{20}{15} = \frac{4}{3}$

$\phi = 53°$

$\tan 53° = \frac{4}{3} = \frac{XY}{400}$

$XY = \frac{1600}{3}$

$AY = \frac{400}{\cos 53°} = \frac{2000}{3}$ m

$\therefore r_{min} = \frac{100}{3} \sin 37°$

$= \frac{20}{\cancel{100}} \times \frac{3}{\cancel{5}} = 20$m

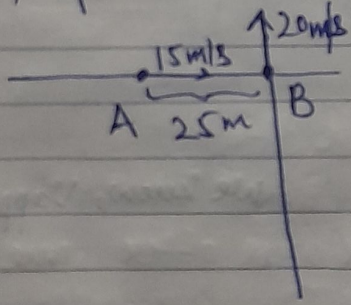
$t_{min sep} = \frac{640}{25} = 25.6$ s

Alternate:

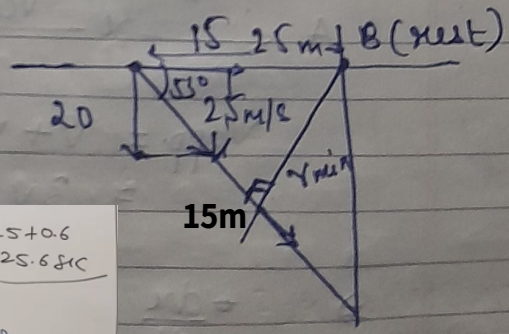
Time taken for A to reach the intersection = $400/15 = 26.6$ s

Time taken for B to reach the intersection = $500/20 = 25$ s

So, after $t = 25$ s,



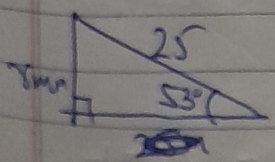
Wrt B



$t = 25 + 0.6 = 25.6$ sec

15m

$\frac{15m}{25 m/sec} = \Delta t = 0.6$ sec



$\sin 53° = \frac{4}{5} = \frac{r_{min}}{25} \Rightarrow r_{min} = \frac{4}{\cancel{5}} \times \cancel{25} = 20$m

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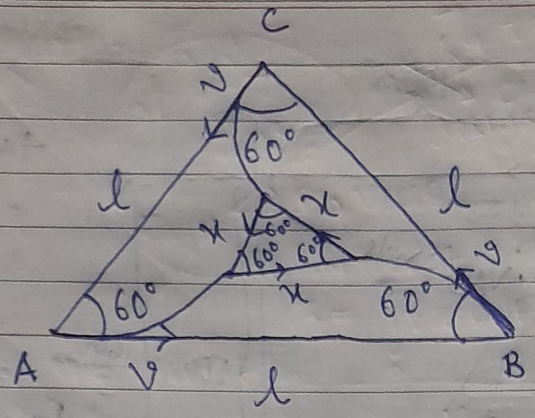
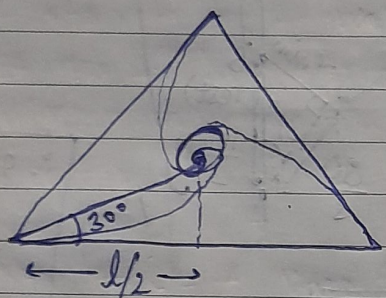
Q



V = constant speed

Find
(a) where will they meet.
(b) Time when they will meet
(c) displacement of any one of them till they meet
(d) distance travelled by any of them till they meet.

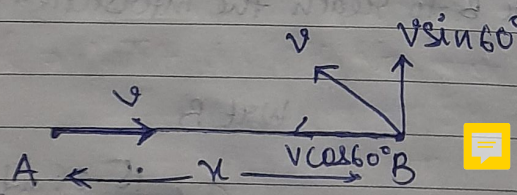
(a)



They will meet at centroid of the triangle (the most symmetric point of the triangle)

(c) displacement = ~~$\frac{l}{2}\cos 30°$~~ $\frac{l}{2\cos 30°} = \frac{l}{\sqrt{3}}$ m

(b)



$v_{app} = v + v\cos 60°$

$= \frac{3v}{2}$

we know, $v_{app} = -\frac{dx}{dt}$

$\Rightarrow -\frac{dx}{dt} = -\frac{3v}{2}$

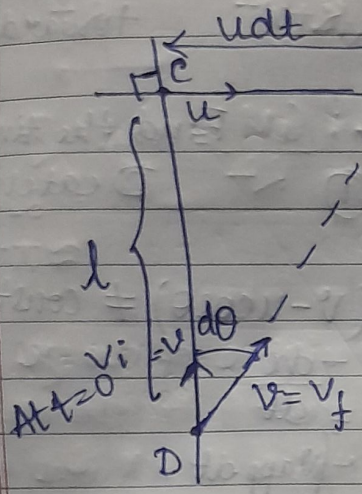
$\Rightarrow \int_0^l dx = -\frac{3v}{2}\int_0^t dt \Rightarrow +l = +\frac{3vt}{2}$

$\boxed{t = \frac{2l}{3v}}$

we could have done without calculus too
as velocity of approach is constant, hence, we have to cover a separation of l with that const. velocity
hence, t = l/Vapp.

(d) $d = \frac{\cancel{v} \times 2l}{3\cancel{v}} \Rightarrow \boxed{d = \frac{2l}{3}}$

Q



C has const. velocity = u
D has const. speed = v
but velocity of D is always directed towards C.
Find initial accn of D.

$$a = \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{|\vec{v_f} - \vec{v_i}|}{dt} = \frac{\sqrt{v^2 + v^2 - 2v^2\cos d\theta}}{dt}$$

$$= \frac{\sqrt{2v^2 - 2v^2\cos d\theta}}{dt} = \frac{v\sqrt{2(1-\cos d\theta)}}{dt} \quad \left(1-\cos\theta = 2\sin^2\frac{\theta}{2},\ 1-\cos d\theta = 2\sin^2\frac{d\theta}{2}\right)$$

$$= \frac{v\sqrt{2\left(2\sin^2\left(\frac{d\theta}{2}\right)\right)}}{dt}$$

we know $\sin x \approx x$ if $x \to 0$

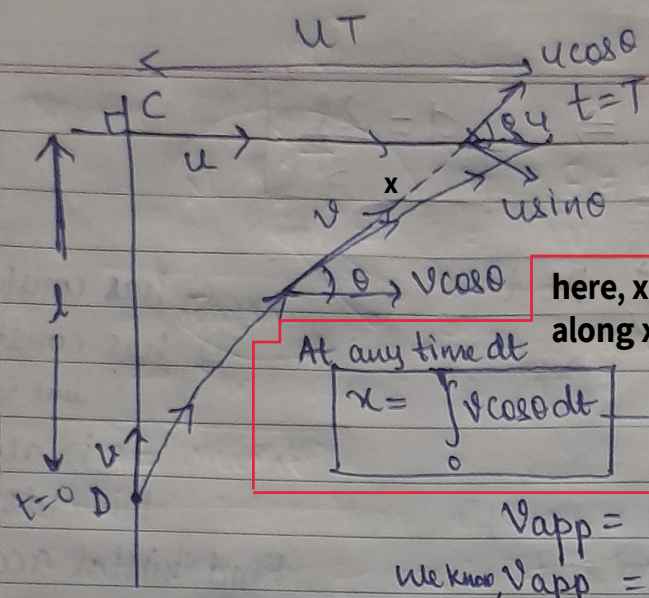
$$= \frac{v(2\sin\frac{d\theta}{2})}{dt} = \frac{v\,\cancel{2}\times\frac{d\theta}{\cancel{2}}}{dt} = \frac{v\,d\theta}{dt}$$

$$a = v\frac{d\theta}{dt} = \frac{v \times \left(\frac{u\,dt}{l}\right)}{dt} = \frac{vu}{l} \qquad \therefore \boxed{a = \frac{uv}{l}}$$

only responsible for changing the direction since speed remains constant

only at initial instant at some other time, the acceleration will be different

Q



Velocity of D is always directed towards C.

Find the time when D catches C

here, x means along x-axis

At any time dt

$$x = \int_0 v\cos\theta\, dt \quad \text{---(1)}$$

$$v_{app} = (v - u\cos\theta) \neq \text{const}$$

we know, $v_{app} = -\frac{dx}{dt}$

here, x means the separation b/w them

$$\int_l^0 dx = -\int_0^t v_{app}\, dt$$

$$= \neq l = \neq \int_0^t v_{app}\, dt \Rightarrow l = \int_0^T v_{app}\, dt$$

$$\Rightarrow l = \int_0^T (v - u\cos\theta)\, dt = v\int_0^T dt - u\int_0^T \cos\theta\, dt$$

$$\Rightarrow l = vT - u\int_0^T \cos\theta\, dt \quad \text{---(2)}$$

displacement of the dog along the x-axis direction

Now from (1) $x = \int_0^T v\cos\theta\, dt$

Total displacement of C° in X axis after time T is

$$x = uT$$

$$\therefore uT = v\int_0^T \cos\theta\, dt$$

$$\Rightarrow \boxed{\int_0^T \cos\theta\, dt = \frac{uT}{v}} \quad \text{---(3)}$$

(1) (2) (3) (4) (I)

Put (3) in (2)

$$l = vT - u\left(\frac{uT}{v}\right) = vT - \frac{u^2T}{v} = T\left(\frac{v^2-u^2}{v}\right)$$

$$\Rightarrow \boxed{T = \frac{lv}{v^2-u^2}}$$

$$S_c = d_c = u \cdot T = \frac{uvl}{v^2-u^2}$$

$$S_D = \sqrt{l^2 + \left(\frac{uvl}{v^2-u^2}\right)^2}$$

$$d_D = v \cdot T = \frac{v^2 l}{v^2-u^2}$$

Newtons

Force: Any kind of pull

→ It has tendency to change

→ It can bring any part

vice versa.

→ It can change the speed

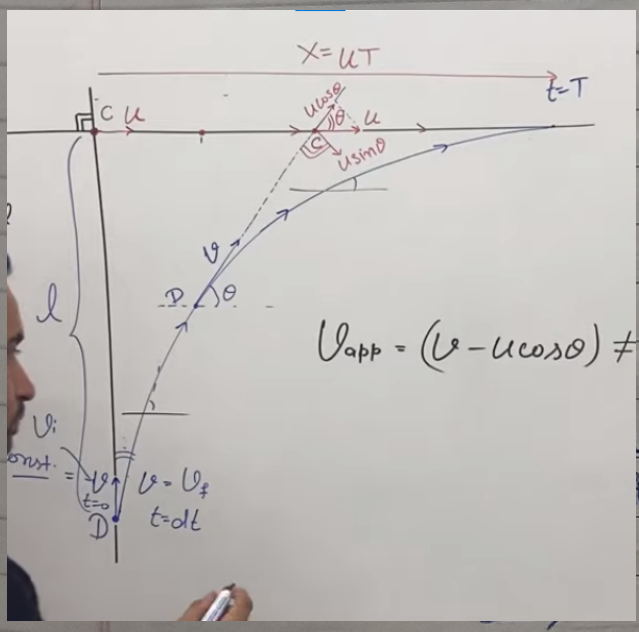
city.

→ It can deform any body.

→ Unit of force = kgm/sec^2 = Newton (N)

In CGS; $1N = \frac{1\,kgm}{sec^2} = \frac{10^3 gm\ 10^2 cm}{sec^2} = 10^5\left(\frac{gm\,cm}{sec^2}\right) = 10^5$ dyne

→ Dimension = $[MLT^{-2}]$



$v_{app} = (v - u\cos\theta) \neq$

} Nuclear force
⇒ Acts in nucleus ⇒ Short Range

sses
lue

$m_1 \longleftrightarrow m_2$, r

$$F = \frac{Gm_1m_2}{r^2}$$

$$G = 6.67 \times 10^{-11}\ \frac{Nm^2}{kg^2} = \frac{20}{3} \times 10^{-11}\ \frac{N}{k}$$